

EvolveIT Tech Company Network Design (Project 3)



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Subnet List

Network Name	No of Hosts	Network Mask	Network Address	1 st Address	Last Address	Broadcast Address
Site1-Marketing	200	255.255.255.0	192.168.1.0	192.168.1.1	192.168.1.254	192.168.1.255
Site1-Operations	35	255.255.255.192	192.168.2.0	192.168.2.1	192.168.2.62	192.168.2.63
Site1-Management	12	255.255.255.240	192.168.3.0	192.168.3.1	192.168.3.14	192.168.3.15
S1Annex-R&D	25	255.255.255.224	192.168.6.0	192.168.6.1	192.168.6.30	192.168.6.31
S1Annex-R2 to Site1-R1	2	255.255.255.252	192.168.7.252	192.168.7.253	192.168.7.254	192.168.7.255
DMZ	5	255.255.255.192	192.168.4.0	192.168.4.1	192.168.4.62	192.168.4.63
Site1-R1 to NETSEC-ASA	2	255.255.255.0	192.168.5.0	192.168.5.1	192.168.5.254	192.168.5.255
Site2-R1 to ASA1	2	255.255.255.0	192.168.11.0	192.168.11.1	192.168.11.254	192.168.11.255
Site2-Business	86	255.255.255.0	192.168.50.0	192.168.50.1	192.168.50.254	192.168.50.255
Site2-Management	86	255.255.255.0	192.168.10.0	192.168.10.1	192.168.10.254	192.168.10.255

Installation/Configuration

1. Install Wireshark

- a. Download 64-bit (For Windows 64-bit version) version file from www.wireshark.org to install in the PCs.
- b. Double clicks the file and start the installation process.
- c. Follow up the steps and if you already have a copy of Wireshark Click **Yes** to uninstall and previous version.
- d. In Wireshark Setup Wizard Click **Next**
- e. Click **I Agree** when the License Agreement Window displays.
- f. On the Choose component window, keep the default settings and Click **Next**
- g. Choose your desired shortcut options and click **Next**
- h. On the installation location window, keep the original location and click **Next**
- i. To capture live network data, WinPcap must be installed on your PC. If WinPcap is already installed on your PC, the Install check box will be unchecked. If you have an older version recommended that you allow the newer version to be installed by clicking the Install WinPcap x.x.x (version number) check box.
- j. Do not install USBPcap for normal traffic capture. Do Not Select the checkbox to install USBPcap (It is experimental).

- k. Wireshark starts installing its files and displays with the status of the installation. Click **Next** when Wireshark installation is complete and then click **Finish** to complete the process.
- l. In a separate window, accept the license agreement in the Npcap Setup Wizard if installing Npcap. Click **I Agree** to continue and then **Next** to finish the Npcap installation and click **Finish** to exit the Npcap installation.

2. Install Putty

- a. In the web browser, go to ninite.com
- b. Under the Development Tools, select Putty
- c. Click **Get your Ninite** button at the bottom of the window.
- d. After downloading, open the installer and run it.

3. Install TFTP server

- a. Go to www.solarwinds.com/free-tools/free-tftp-server
- b. Click **Download free tool**
- c. First Name: Hashan Last Name: Kodippilige
- d. Business Email: hashan@hsk.com
- e. Company: EvolveIT Tech
- f. Zip Code: 58203
- g. Business Phone: 7014108095
- h. Install
- i. Continue without adding
- j. Windows-Download now
- m. After you download, double click the file and start the installation process.

4. Saving configurations to TFTP server after each Session

4.1 Switches (Do it on each switch) and Router

Switches	Router (Site1-R1)
Site1-S1# copy flash:vlan.dat tftp Destination IP: 192.168.3.2 File name: s1-vlan.dat Save the configuration	Site1-R1# copy running-config tftp Destination IP: 192.168.3.2 File name: r1-config Save the configuration
Site1-S1# copy running-config tftp Destination IP: 192.168.3.2 File name: s1-config Save the configuration	Site1-R1# erase startup-config
Site1-S1# delete vlan.dat Site1-S1# erase startup-config	

5. Restoring configurations from TFTP server

5.1 Switches (Do it for each switch) and Router

Switches	Router (Site1-R1)
switch# copy tftp flash:vlan.dat (Answer questions) File name: s1-vlan.dat switch # copy tftp running-config (Answer questions) File name: s1-config Save the configuration	Site1-R1# copy tftp: running-config tftp (Answer questions) File name: r1-config Save the configuration

Devices and Connections

Device Ports	Connected to Device Ports	Purpose
Site1-R1: G0/0/1	Site1-S1:F0/5	Trunk
Site1-R1: G0/0/0	ASA Firewall G1/2	
Site1-S1: F0/1-2	Site1-S2: F0/1-2	Trunk
Site1-S4: F0/6-17	Management Group	VLAN 20
Site1-S1: F0/6-14	Operations Group	VLAN 30
Site1-S1: F0/15-24	Marketing Group	VLAN 40
Site1-S3:F0/1	ASA Firewall G1/3	Uplink to ASA Firewall
ASA Firewall: G1/1	ISP	
Site1-S3:F0/6	Web Server	VLAN 50
Site1-S2:F0/3-4	Site1-S4:F0/1-2	Trunk
Site1-S1:F0/3	Site1-S4:F0/3	Trunk
Site2-R1: G0/0/0	Site2-S1: F0/22	Trunk
Site2-R1: G0/0/1	ASA Firewall G1/1	
Site2-S1: G0/1-2	Site2-S2: G0/1-2	Trunk
Site2-S2: F0/1	Management Group	VLAN 10
Site2-S1: F0/1-20	Business Group	VLAN 50
Site2-S2: F0/2-24	Operations Group	VLAN 50
Site2-S1: F0/21, 23-24	Business Group	VLAN 50
S1Annex-R2:S0/1/0	Site1-R1:S0/1/0	
S1Annex-R2:G0/0/1	S1Annex-S5:F0/5	VLAN 60
S1Annex-S5:F0/6-24	R&D Group	VLAN 60

Basic Configurations for Network Devices (See Appendix -1)

Device	Interface	IP Address	Subnet Mask	Default Gateway
Site1-R1	G0/0/0	192.168.5.253	255.255.255.252	N/A
	G0/0/1	N/A	N/A	
	G0/0/1.20	192.168.3.1	255.255.255.240	
	G0/0/1.30	192.168.2.1	255.255.255.192	
	G0/0/1.40	192.168.1.1	255.255.255.0	
	G0/0/1.1000	N/A	N/A	
	S0/0/0	192.168.7.254	255.255.255.252	
S1Annex-R2	G0/0/1	192.168.6.1	255.255.255.224	N/A
	S0/1/0	192.168.7.253	255.255.255.252	
Site2-R1	G0/0/0	N/A	N/A	N/A
	G0/0/0.10	192.168.10.1	255.255.255.0	
	G0/0/0.50	192.168.50.1	255.255.255.0	
	G0/0/0.1000	N/A	N/A	
	G0/0/1	192.168.11.1	255.255.255.0	
NETSEC-ASA	G1/1 (OUTSIDE)	172.24.104.4	255.255.255.248	N/A
	G1/2 (INSIDE)	192.168.5.254	255.255.255.252	
	G1/3 (DMZ)	192.168.4.1	255.255.255.192	
ASA1 (Site2)	G1/1 (INSIDE)	192.168.11.2	255.255.255.0	N/A
	G1/3 (OUTSIDE)	172.24.160.4	255.255.255.0	
Site1-S1	VLAN 30	DHCP	255.255.255.192	192.168.2.1
Site1-S2	VLAN 40	DHCP	255.255.255.0	192.168.1.1
Site1-S3	VLAN 50	192.168.4.2	255.255.255.192	192.168.4.1
Site1-S4	VLAN 20	DHCP	255.255.255.240	192.168.3.1
Site2-S1	VLAN 10	192.168.10.2	255.255.255.0	192.168.10.1
Site2-S2	VLAN 10	192.168.10.3	255.255.255.0	192.168.10.1
S1Annex-R2	G0/0/1	N/A	N/A	N/A
	G0/0/1.60	192.168.6.1	255.255.255.224	
	G0/0/1.1000	N/A	N/A	
	S0/1/0	192.168.7.254	255.255.255.252	
S1Annex-S5	VLAN 60	192.168.6.2	255.255.255.224	192.168.6.1
TFTP Server	NIC	192.168.3.2	255.255.255.240	192.168.3.1
Web Server	NIC & ens33 -Ethernet	192.168.4.3 & 192.168.4.4	255.255.255.192	192.168.4.1
	ens33.0 -Intranet	192.168.4.45	255.255.255.192	192.168.4.1
	ens33.1 -Internet	192.168.4.46	255.255.255.192	192.168.4.1
	ens33.2 -B2B	192.168.4.47	255.255.255.192	192.168.4.1
Business Server	NIC	192.168.2.2	255.255.255.192	192.168.2.1
Local Server (Site2)	NIC	192.168.20.2	255.255.255.192	192.168.20.1
S1Annex-PC(5.1-5.25)	NIC	192.168.6.3 – 192.168.6.31	255.255.255.224	192.168.6.1

Site1-PC(3.1-3.11)	NIC	192.168.3.4 – 192.168.3.14	255.255.255.240	192.168.3.1
Site1-PC-(2.1-2.34)	NIC	192.168.2.4 – 192.168.2.37	255.255.255.192	192.168.2.1
Site1-PC-(1.1-1.200)	NIC	192.168.1.5 – 192.168.1.204	255.255.255.0	192.168.1.1
Site2-PC(10.1-10.86)	NIC	192.168.10.4 – 192.168.10.254	255.255.255.0	192.168.10.1
Site2-PC-(50.1-50.86)	NIC	192.168.50.3 – 192.168.50.254	255.255.255.0	192.168.50.1

1. Router

- Assign a device name to the router.
- Disable DNS lookup to prevent the router from attempting to translate incorrectly entered commands as though they were host names.
- Assign class as the privileged EXEC encrypted password.
- Assign cisco as the console password and enable login.
- Assign cisco as the VTY password and enable login.
- Encrypt the plaintext passwords.
- Create a banner that warns anyone accessing the device that unauthorized access is prohibited.
- Save the running configuration to the startup configuration file.

2. ASA Firewall

- Assign a device name to the firewall.
- Assign domain name as hsk-spare.com
- Assign cisco as login password.
- Create a banner that warns anyone accessing the device that unauthorized access is prohibited.
- Save the running configuration to the startup configuration file.

3. Switches

- Assign a device name to the switch.
- Disable DNS lookup to prevent the router from attempting to translate incorrectly entered commands as though they were host names.
- Assign class as the privileged EXEC encrypted password.
- Assign cisco as the console password and enable login.
- Assign cisco as the VTY password and enable login.
- Encrypt the plain text passwords.
- Create a banner that warns anyone accessing the device that unauthorized access is prohibited.
- Save the running configuration to the startup configuration file.

4. PCs

- Assigning PCs host address will be done through DHCP server and Assign TFTP server, Business server., Local server and Web server addresses as above table.

List of Each Feature Configured (See Appendix -2)

VLAN	Name	Interface Assigned
20	Management	Site1-S4: F0/6-17
10	Management	Site2-S1: F0/21, 23-24 Site2-S2: F0/1
30	Operations	Site1-S1: F0/6-14
50	Business	Site2-S1: F0/1-20 Site2-S2: F0/2-24
40	Marketing	Site1-S1: F0/15-24
50	DMZ	Site1-S3: F0/1 and F0/6
60	R&D	S1Annex-S5: F0/5, F0/6-24
999	ParkingLot	Site1-S1: F0/4, G0/1-2 Site1-S2: F0/5-24, G0/1-2 Site1-S3: F0/2-5, F0/7-24, G0/1-2 Site1-S4: F0/4-5, F0/18-24, G0/1-2 S1Annex-S5: F0/1-4, G0/1-2
1000	Native	N/A (Site1)
2	Native	N/A (Site2)

1. Create VLANs on seven switches

- Create VLAN as shown in the table above
- Configure the management interface and default gateway on each switch using the IP address information in the Addressing Table.
- Assign used ports to the appropriate VLAN (specified in the VLAN table above) and configure them for static access mode.
- Assign all unused ports on the switch to the Parking Lot VLAN, configure them for static access mode, and administratively deactivate them.
- Verify configuration by issuing **Show Vlan Brief** command.

2. Configure Trunking

2.3 Manually configure trunk interface F0/1-2 on Site1-S1 and S2, and F0/3-4 on Site1-S2 and F0/1-2 on S4, and Interface G0/1-2 on Site2-S1 & S2 Switches with EtherChannel

- Specify the interfaces using interface range command.
- Create the port channel interface with the channel-group identifier mode active command.
- To change the layer 2 settings on the port channel interface, enter port channel interface configuration mode using interface port-channel command.
- Change the switchport mode on interface to force trunking. Do it on both switches.
- Set up Native vlan to 1000 on both switches.
- Specify that Vlan 20,30,40 and 1000 are allowed to cross the trunk.

- g. Verify configuration by issuing **Show interface trunk** command.

2.2 Manually configure Site1-S1 trunk interfaces F0/3, and F0/5. Site1-S4 trunk interface F0/3. Site2-S1 trunk interfaces F0/22.

- a. Use the same configuration as F0/1-2 on Site1-S1 switch to configure trunking on interface F0/5 on Site1-S1 and interface F0/22 on Site2-S1, interface F0/3 on both Site1-S1&S4.
- b. Save configuration from Running to Startup.
- c. Verify configuration by issuing **Show interface trunk** command.

3. PortFast and BPDU Guard Configuration on Switches (Do not configure on Site1-S1&S4 interface F0/3.

- a. It can be done either globally or on a per-interface basis. When enable globally, portfast is enabled on all non-trunking ports. Enter the global configuration mode.
- b. Enter **spanning-tree portfast default** command on the global configuration mode.
- c. Enter **spanning-tree bpduguard enable** (Can do it on both portfast and bpduguard in the same command line using **spanning-tree portfast bpduguard default** command)

4. Configure Inter-Vlan Routing on both Routers (Site1-R1 and Site2-R1).

- a. Activate interface G0/0/1 on Site1-R1 router and G0/0/0 on Site2-R1 router.
- b. Configure Sub-interfaces for each VLAN as specified in the IP addressing table. All sub-interfaces use 802.1Q encapsulation. Do not configure IP address for Native VLAN. Include description for each sub-interface.
- c. Verify configuration by issuing **Show ip interface brief** command.

5. Configure Default Routing for all Routers

- a. Configure G0/0/0 and G0/0/1 to IP address given on the table on Site1-R1, & Site2-R1, and Configure S0/1/0 to IP address given on table for S1Annex-R2 and Site1-R1.
- b. Default route from Site1-R1 to G1/2 interface on NETSEC-ASA interface and from Site2-R1 to G1/2 interface G1/1 interface on ASA1, and S1Annex-R2 to S0/1/0 interface on Site1-R1.

6. Configure DHCP on all Routers

- a. Exclude statically assigned addresses for all vlans separately.
- b. Configure the pool name with Vlan name and specify the subnet to use when assigning IP addresses.
- c. Configure dns server
- d. Configure the default router and domain name server for the network.
- e. Verify the DHCP configuration.

7. Configure PAT on Routers (Site1-R1)

- a. Define the access list for internal network
- b. Configure the outside interface
- c. Configure inside interfaces
- d. Enable NAT overload (PAT)

8. Configure all network devices for basic SSH support (All Routers, and all Switches).

- a. Create a local user with username *admin* and on Site2 username as *admin01* and encrypted password *cisco12345*
- b. Use hsk-spare.com as domain name.
- c. Generate crypto keys using a 2048-bit modulus.
- d. Configure the first five VTY lines on each device to support SSH connections only and to authenticate to local user database.
- e. Issue the command to force the use of SSH version 2.

9. Configure both ASA interfaces

- a. Configure the OUTSIDE interface
 - Assign IP address, name the interface, and set the security level to 0 (least trusted)
- b. Configure the INSIDE interface
 - Set the IP address and the highest security level (100) for internal LAN access
- c. Configure the DMZ interface on NETSEC-ASA.
 - Assign an IP address, name the interface, and set the security level to 70

10. Configure Default route for the ASA

- a. Set static default route to the next hop (ISP) and static route to the INSIDE network

11. Configure address translation using PAT and network objects

- a. Create a network object INSIDE-NET
 - Assign attribute to it using the *subnet* and *nat* command.
- b. Create a network object DMZ-NET on NETSEC-ASA
 - Assign attribute to it using the *subnet* and *nat* command.

12. Enable ICMP Inspection in Global Policy

- a. Add the inspection of ICMP traffic to the policy map.
 - Check policy map using *show run policy-map* command

13. Configure Static NAT to the DMZ Server using a network object on Site1 ASA

- a. Create a network object for the Internet website
 - Define static NAT rules to map it to the outside Public IP

14. Configure an ACL to allow access to the Web server from the Internet and all servers to access from INSIDE network

- a. Allow OUTSIDE-network to access the Internet website.
 - Allow OUTSIDE access to Internet server (static NAT: 172.24.104.7 to 192.168.4.46)
 - Apply the ACL to the ASA OUTSIDE interface in the *in* direction.
- b. Apply the ACL to the ASA to any DMZ host to access any INSIDE host through ping.
 - Apply the ASA any DMZ host to access any INSIDE host through tcp on HTTP and HTTPS.
 - Apply the ACL to the ASA DMZ interface in the *in* direction.

15. Configure AAA to use local database for authentication and SSH remote access to ASA

- a. Create a local user named *admin* for Site1 Asa and *admin01* for Site2-ASA and password *cisco12345*
- b. Configure AAA to use the local ASA database for SSH user authentication.
- c. Generate an RSA key as 2048 and save the RSA key to persistent flash memory
- d. Configure the ASA to allow SSH connections from any host from Inside network
- e. Set SSH timeout for 10 minutes.

16. Configure Site-to-Site VPN on both ASA Firewalls (Same Configuration for both ASA)

- a. Configure object-group for SITE1-NET and SITE2-NET by including all networks attached to their site.
- b. Configure the IKEV2 policies on ASAs.
 - Configure an IKEV2 policy with a priority of 10.
 - Use *pre-shared key* as the *authentication* type, *aes 256* for the encryption algorithm
 - *sha* as the *hash* algorithm, and the Diffie-Hellman group 2 key exchange.
 - Give a policy a lifetime of 86400 seconds (One hour)
- c. Enable IKEv1 on the interface that terminates the VPN tunnel (OUTSIDE interface).
- d. Define the IPsec Transform Set
 - Named transform set as VPN-SET and specifies 256-bit AES encryption and SHA-1 HMAC for digital integrity.
- e. Configure Interesting Traffic and Crypto ACL.
 - The extended ACL named VPN-TRAFFIC. Define traffic between two object groups SITE1-NET and SITE2-NET.
 - Permit VPN traffic from INSIDE network.
 - Permit IKE/IPsec from OUTSIDE and allow UDP eq 500(ISAKMP), 4500 and ESP (Encapsulated Security Payload) for OUTSIDE-DMZ.
 - Permit Site2 to access DMZ
 - Permit INSIDE-IN network to access IP any any.
- f. Configure the Tunnel Group with Pre-shared keys.
 - Connection profile type is IPSec-121
 - Configure profile attribute with pre-shared key of cisco123 with ASA OUTSIDE interface IP addresses.
- g. Configure a Crypto Map and Bind it to the Interface.
 - Configure previously configured access list to match the traffic that must be encrypted.
 - Sets the IP of the remote VPN device (Peer).

- Configure the peer identification IKEv1 IPSec proposal.
- Apply crypto map VPN-MAP to the ASA OUTSIDE interface.
- h. Configure an Identity NAT.
 - Configure static NAT for INSIDE to OUTSIDE network.
 - Configure static NAT to DMZ to OUTSIDE network.
- i. Configure route on routers for VPN connection

17. Configure OSPF on Routers

- a. Configure OSPF process ID as 101 for all routers.
- b. Configure router ID
- c. Configure *ospf networks* and wildcard masks relevant to each router and assign *Area 0* for all the routers and ASA.

18. Configure Business Partners to access the B2B web.

- a. Create a static NAT for B2B net.
- b. Create ACL to access business partners from OUTSIDE interface.

19. Configure NTP in the Network

- a. Set the clock and configure Site1-R1 as NTP master 4.
- b. Configure NTP server on ASA with source inside command
- c. Configure NTP on S1Annex-R2, Site2-R1, and switches

20. Configure the Web server

- a. Download and install updated VMware on the webserver machine.
- b. Create a new virtual machine with Ubuntu install as operating system.
- c. Install webmin to the virtual machine.
- d. Go to Network > Interfaces and create new virtual interfaces with eth0 – eth3 and assign ip address , subnet mask, and broadcast address as shown in the table.
- e. Test access to the web server

21. Verify the Network

From	Protocol	Destination	Result
Site1-PC-1.1	Ping	Site1-PC-50.1	Success
Web server	Ping	192.168.5.254	Success
Site1-PC-1.1	HTTP	192.168.4.45-47	Success
Site1-PC-1.1	Ping via VPN	Site2-PC-10.1	Success
Site1-PC-1.1	SSH	192.168.1.1	Success
Site1-PC-1.1	SSH	192.168.5.254	Success
Site1-PC-1.1	SSH	192.168.6.1	Success
Site2-PC-10.1	HTTP	192.168.4.45-47	Success
Site2-PC-10.1	Ping	Site1-PC-1.1	Success
Business Partner	HTTP	172.24.104.5	Success
Internet Users	HTTP	172.24.104.6	Success

22. Troubleshooting

a. R&D users do not have connection.

- Did show run command on S1Annex-S5 switch and found that interface F0/5 configured into ParkingLot Vlan, and DHCP pool does not configure on S1Annex-R2.

```
S1Annex-S5(config)# interface f0/5
S1Annex-S5 (config-if)# switchport mode access
S1Annex-S5 (config-if)# switchport access vlan 60
S1Annex-S5 (config-if)# exit
```

```
S1Annex-R2(config)#ip dhcp excluded-address 192.168.6.1 192.168.6.2
S1Annex-R2(config)#ip dhcp pool R&D
S1Annex-R2(config-dhcp)# network 192.168.6.0 255.255.255.224
S1Annex-R2(config-dhcp)# dns-server 8.8.8.8
S1Annex-R2(config-dhcp)# default-router 192.168.6.1
S1Annex-R2(config-dhcp)# domain-name hsk-spare.com
```

b. Configure VPN connection on NETSEC-ASA and Site2-R1

- Face numerous problems with commands and configuring VPN with a Router and ASA. Also, the best practice is to keep VPN connection between two Firewalls. Choose to configure VPN on NETSEC-ASA and ASA1 due to security on Site2, If I configure VPN on Site2-R1, there is no security measures on Site2 without a Firewall.

c. Configuring VPN connection on NETSEC-ASA and ASA1 using IKEv2

- First tried to configure ikev2 configuration on firewalls.
- It has numerous problems establishing VPN connection with IKEv2 commands, tried to troubleshoot the problems with my ability but some commands cannot configure on ASAs.

Choose to configure ikev1 configuration on Firewalls.

d. Configuring VPN connection on NETSEC-ASA and ASA1 using IKEv1

- VPN connection not established between two ASAs.
- Use *show crypto isakmp sa* and *show crypto ipsec sa* command to check the status of the VPN connection. Connection status is Drop.
- Check the NAT exemption rules using *show run nat* command and found No NAT exemption rules configure on both ASA interfaces. Configured NAT config on both ASAs.

```
NETSEC-ASA(config)# nat (INSIDE,OUTSIDE) source static SITE1-NET SITE1-
NET destination static SITE2-NET SITE2-NET no-proxy-arp
NETSEC-ASA(config)# nat (DMZ,OUTSIDE) source static SITE1-NET SITE1-NET
destination static SITE2-NET SITE2-NET no-proxy-arp
```

```
ASA1(config)# nat (INSIDE,OUTSIDE) source static SITE2-NET SITE2-NET
destination static SITE1-NET SITE1-NET no-proxy-arp
```

- Then my VPN status is *MM_ACTIVE*, but still not fully configured VPN. Clear tunnels using *clear crypto ikev1 sa* and *clear crypto ipsec sa* commands, and then enable debug using *debug crypto ikev1* and *debug crypto ipsec* commands on both ASAs. But still *MM_ACTIVE* with *Rekey: no*

Reconfigure ASAs with crypto key and now I can ping from Site1 to Site2 machines. But still *MM_ACTIVE* (IKEv1 Phase 1 negotiation is in progress - Main Mode is active, but not yet completed).

Use *packet-tracer input inside icmp 192.168.3.4 0 0 192.168.10.6* command on Site1

Phase 9 VPN : drop. Double check the Crypto ACL are matching on both ASAs using *show access-list VPN-TRAFFIC* command and it also matched.

Allow all INSIDE traffic and configured ip route command on routers.

```
NETSEC-ASA(config)# access-list INSIDE-IN extended permit ip any any
NETSEC-ASA(config)# access-group INSIDE-IN in interface INSIDE
ASA1(config)# access-list INSIDE-IN extended permit ip any any
ASA1(config)# access-group INSIDE-IN in interface INSIDE
```

```
Site1-R1(config)# ip route 192.168.10.0 255.255.255.0 192.168.5.254
Site1-R1(config)# ip route 192.168.11.0 255.255.255.0 192.168.5.254
Site1-R1(config)# ip route 192.168.50.0 255.255.255.0 192.168.5.254
Site1-R1(config)# ip route 192.168.6.0 255.255.255.0 192.168.7.253
```

```
Site2-R1(config)# ip route 192.168.1.0 255.255.255.0 192.168.11.2
Site2-R1(config)# ip route 192.168.2.0 255.255.255.192 192.168.11.2
Site2-R1(config)# ip route 192.168.3.0 255.255.255.240 192.168.11.2
Site2-R1(config)# ip route 192.168.4.0 255.255.255.192 192.168.11.2
Site2-R1(config)# ip route 192.168.5.0 255.255.255.0 192.168.11.2
Site2-R1(config)# ip route 192.168.6.0 255.255.255.192 192.168.11.2
```

- Pings are working only on Site1 to Site2 network, Site2 to Site1 not working ping 192.168.3.5 from Site2, but it fails.
tracert 192.168.3.5 from Site2-PC and it only reach to next hop 192.168.50.1
- Use *show access-list* and *show access-group* command on both ASAs to find the differences with Access-lists.
- The final items that were unable to be configured were the following:
 1. Fully functional site to site VPN. Me and Travis think that it is functional and working properly there is most likely an ACL or routing function that is preventing access to the inside network of site 1 from site 2
 2. NTP was not able to be fully implemented. We set out master server in Site1-R1, about half the switches in site 1 were able to connect to the NTP server and get their

time form it. The other half were set manually (Site1-S3 and Site1-S4) as close to the NTP as we could.

23. Daily Journal

Over the weekend I prepared the document with all configurations needed to implement the Project Phase3. I documented all required VLAN, connected interfaces and the configurations for all the devices.

- a. **5/5/2025** – I and the other network engineer started to implement the network according to our network plan. After we set up the basic configurations for Site1 and Site2, we started with multiple attempts to configure a site-to-site VPN between our equipment. First, we tried to configure VPN between two sites using Site1 ASA and Site2-R1.

Steps Taken:

1. Establish Site1 and Site2 basic configurations.
2. Decommission webserver and switch 3 from site 2 topology
3. Attempt #1 at creating a site-to-site VPN using Router and ASA firewall.

- b. **5/6/2025** – Today me and my network engineer partner spend again most of the day attempting to set up the site-to-site VPN. We rolled back our configurations and tried NETSEC groups one more time before deciding to shift focus for the rest of the day and configure and physically set up a new annex location named R&D. This location was set up with a serial connection. A new router and switch would need to be set up and configured along with a new pc for the R&D lab.

Steps Taken:

1. Added a new router S1Annex-R2 connected to Site1-R1 via a serial connection.
2. Configured a switch in the new annex zone.
3. Attempt #2 of configuring site-to-site VPN.

- c. **5/7/2025** – The day started off strong with me and my partner deciding to go a different path with the site-to-site VPN. After some research we changed our approach from router to ASA VPN to a new method, ASA to ASA VPN. This would also prove to be difficult and took some time and many rollbacks to get going.

Steps Taken:

1. Continued configuration on ASAs for VPN connection

- d. **5/8/2025** – Today was spend configuring and trying out Ikev1 instead of Ikev2 on our ASAs to get our site-to-site VPN working. It was also discovered that our R&D Department annex was not receiving internet access. We divided up so that one of us was working on the ASA/VPN and the other was investigating and correcting the misconfiguration of the R&D annex.

Steps Taken:

1. Attempt #3 of the site-to-site vpn
2. DHCP and VLAN configs set up on S1Annex equipment.

- e. **5/12/2025** – Final touches and troubleshooting are being done as we wrap up the project. I estimate with our knowledge and timeline we were able to complete 85%-90% of the

requested features. The rest of the features would be able to come online if we had more time to meet the deadline.

Steps Taken:

1. Final annex configuration was made.
2. Final Attempt at a functioning site to site VPN was made. We were not successful.

- f. **5/13/2025** – Try to figure out what happened with Site1 ACL but couldn't figure it out, change all the ACL and configure them again and tried to identify the problem with the VPN connection.

Steps Taken:

1. Tried to configure ACL again and tried to find the problem with VPN connection.

- g. **5/14/2025** – Tried for last time to figure out the root cause for the problem. Use numerous show command to find the differences between the Site1 and Site2 configurations and nothing suspicious were found.

Budget Estimation

Item	Unit Price (\$)	Quantity	Total Price (\$)
Router (ISR4321/K9)	1128.00	03	3384.00
Switch (WS-C2960-24TT-L)	625.00	07	4375.00
Firewall (ASA 5506-X)	758.78	02	1517.56
Web Server (Beelink Sei8 Mini PC4 Core 8 th Gen intel i5-8279U)	473.83	01	473.83
Pcs	749.99	419	314,245.81
Printer (Brother MFC-L3780CDW Color Multifunction Printer)	529.00	05	2645.00
Cables(Cat6, Console)	5.13	1200	6156.00
UPS backup units (1000VA)	245.99	5	1229.95
Rack, Casings and Miscellaneous	-	-	15,000.00
Total Amount			349,026.20

Appendix - 1

Basic Configurations for Network Devices

1. Routers (This is for Site1-R1, configure it for S1Annex-R2, and Site2-R1 also)

- a. router(config)# hostname Site1-R1
- b. Site1-R1(config)# no ip domain lookup
- c. Site1-R1(config)# enable secret class
- d. Site1-R1(config)# line console 0
Site1-R1(config-line)# password cisco
Site1-R1(config-line)# login
- e. Site1-R1(config)# line vty 0 4
Site1-R1(config-line)# password cisco
Site1-R1(config-line)# login
- f. Site1-R1(config)# service password-encryption
- g. Site1-R1(config)# banner motd \$ Unauthorized access is Prohibited! \$
- h. Site1-R1# copy running-config startup-config

2. ASA Firewalls (This is for NETSEC-ASA and configure it for ASA1 also)

- a. ciscoasa(config)# hostname NETSEC-ASA
- b. NETSEC-ASA(config)# domain-name hsk-spare.com
- c. NETSEC-ASA(config)# enable password cisco
- d. NETSEC-ASA(config)# banner motd \$ Unauthorized access is Prohibited! \$
- e. NETSEC-ASA(config)# write memory

3. Switches (This is for Site1-S1 and configure on all switches according to their names)

- a. switch(config)# hostname Site1-S1
- b. Site1-S1(config)# no ip domain lookup
- c. Site1-S1(config)# enable secret class
- d. Site1-S1(config)# line console 0
Site1-S1(config-line)# password cisco
Site1-S1(config-line)# login
- e. Site1-S1(config)# line vty 0 15
Site1-S1(config-line)# password cisco
Site1-S1(config-line)# login
- f. Site1-S1(config)# service password-encryption
- g. Site1-S1(config)# banner motd \$ Unauthorized access is Prohibited! \$
- h. Site1-S1# copy running-config startup-config

4. Assigning of PCs host address will be done through DHCP server and Assign TFTP server, Business server and Web server addresses as below table.

Device	Interface	IP Address	Subnet Mask	Default Gateway
TFTP Server	NIC	192.168.3.2	255.255.255.240	192.168.3.1
Web Server	NIC	192.168.4.3	255.255.255.192	192.168.4.1
Business Server	NIC	192.168.2.2	255.255.255.192	192.168.2.1

Appendix - 2

List of Each Feature Configured

1. a. Create VLANs on switches (This is for Site1-S1 and configure on Site1-S2, S4 same configuration. For Site2-S1 and S2 create VLAN10, 50. For Site1-S3 configure Vlan 50, 999. The Vlan 60, 999 on S1Annex1-S5)

```
Site1-S1 (config)# vlan 20
Site1-S1 (config-vlan)# name Management
Site1-S1(config-vlan)# vlan 30
Site1-S1(config-vlan)# name Operations
Site1-S1(config-vlan)# vlan 40
Site1-S1(config-vlan)# name Marketing
Site1-S1(config-vlan)# vlan 999
Site1-S1(config-vlan)# name ParkingLot
Site1-S1(config-vlan)# vlan 1000
Site1-S1(config-vlan)# name Native
Site1-S1(config-vlan)# exit
```

- b. Site1-S1(config)# interface vlan 30
Site1-S1(config-if)# ip address dhcp
Site1-S1(config-if)# no shutdown
Site1-S1(config-if)# exit
Site1-S1(config)# ip default-gateway
192.168.2.1
Site1-S1(config)# end

```
Site1-S2(config)# interface vlan 40
Site1-S2(config-if)# ip address dhcp
Site1-S2(config-if)# no shutdown
Site1-S2(config-if)# exit
Site1-S2(config)# ip default-gateway
192.168.1.1
Site1-S2(config)# end
```

```
Site1-S3(config)# interface vlan 50
Site1-S3(config-if)# ip address 192.168.4.2
255.255.255.192
Site1-S3(config-if)# no shutdown
Site1-S3(config-if)# exit
Site1-S3(config)# ip default-gateway
192.168.4.1
Site1-S3(config)# end
```

```
Site1-S4(config)# interface vlan 20
Site1-S4(config-if)# ip address dhcp
Site1-S4(config-if)# no shutdown
Site1-S4(config-if)# exit
Site1-S4(config)# ip default-gateway
192.168.3.1
Site1-S4(config)# end
```

```
S1Annex-S5(config)# interface vlan 60
S1Annex-S5 (config-if)# ip address 192.168.6.2
255.255.255.192
S1Annex-S5 (config-if)# no shutdown
S1Annex-S5 (config-if)# exit
S1Annex-S5 (config)# ip default-gateway
192.168.6.1
S1Annex-S5 (config)# end
Site2-S1(config)# interface vlan 10
Site2-S1(config-if)# ip address 192.168.10.2
255.255.255.0
Site2-S1(config-if)# no shutdown
Site2-S1(config-if)# exit
Site2-S1(config)# ip default-gateway
192.168.10.1
Site2-S1(config)# end
```

```
Site2-S2(config)# interface vlan 10
Site2-S2(config-if)# ip address 192.168.10.3 255.255.255.0
Site2-S2(config-if)# no shutdown
Site2-S2(config-if)# exit
Site2-S2(config)# ip default-gateway 192.168.10.1
Site2-S2(config)# end
```

- c. Site1-S1(config)# interface range f0/6 - 14
Site1-S1(config-if-range)# switchport mode access
Site1-S1(config-if-range)# switchport access vlan 30
Site1-S1(config-if-range)# exit
Site1-S1(config)# interface range f0/15 - 24
Site1-S1(config-if-range)# switchport mode access
Site1-S1(config-if-range)# switchport access vlan 40
Site1-S1(config-if-range)# exit

```
Site1-S3(config)# interface range f0/1, f0/6
Site1-S3(config-if-range)# switchport mode access
Site1-S3(config-if-range)# switchport access vlan 50
Site1-S3(config-if-range)# exit
```

```
Site1-S4(config)# interface range f0/6 - 17
Site1-S4(config-if-range)# switchport mode access
Site1-S4(config-if-range)# switchport access vlan 20
Site1-S4(config-if-range)# exit
```

```
S1Annex-S5(config)# interface range f0/5, f0/6-24
S1Annex-S5 (config-if-range)# switchport mode access
S1Annex-S5 (config-if-range)# switchport access vlan 60
S1Annex-S5 (config-if-range)# exit
```

```
Site2-S1(config)# interface range f0/1 - 20
Site2-S1(config-if-range)# switchport mode access
Site2-S1(config-if-range)# switchport access vlan 50
Site2-S1(config-if-range)# exit
Site2-S1(config)# interface range f0/21, 23 - 24
Site2-S1(config-if-range)# switchport mode access
Site2-S1(config-if-range)# switchport access vlan 10
Site2-S1(config-if-range)# exit
```

```
Site2-S2(config)# interface f0/1
Site2-S2(config-if)# switchport mode access
Site2-S2(config-if)# switchport access vlan 10
Site2-S2(config-if)# exit
Site2-S2(config)# interface range f0/2 - 24
Site2-S2(config-if-range)# switchport mode access
```

```
Site2-S2(config-if-range)# switchport access vlan 50
Site2-S2(config-if-range)# exit
```

- d. Site1-S1(config)# interface range f0/4, g0/1 - 2
Site1-S1(config-if-range)# switchport mode access
Site1-S1(config-if-range)# switchport access vlan 999
Site1-S1(config-if-range)# shutdown
Site1-S1(config-if-range)# end

```
Site1-S2(config)# interface range f0/5-24, g0/1-2
Site1-S2(config-if-range)# switchport mode access
Site1-S2(config-if-range)# switchport access vlan 999
Site1-S2(config-if-range)# shutdown
Site1-S2(config-if-range)# end
```

```
Site1-S3(config)# interface range f0/2-5, f0/7-24, g0/1-2
Site1-S3(config-if-range)# switchport mode access
Site1-S3(config-if-range)# switchport access vlan 999
Site1-S3(config-if-range)# shutdown
Site1-S3(config-if-range)# end
```

```
Site1-S4(config)# interface range f0/4-5, f0/18-24, g0/1-2
Site1-S4(config-if-range)# switchport mode access
Site1-S4(config-if-range)# switchport access vlan 999
Site1-S4(config-if-range)# shutdown
Site1-S4(config-if-range)# end
```

```
S1Annex-S5(config)# interface range f0/1-4, g0/1-2
S1Annex-S5(config-if-range)# switchport mode access
S1Annex-S5(config-if-range)# switchport access vlan 999
S1Annex-S5(config-if-range)# shutdown
S1Annex-S5(config-if-range)# end
```

- e. Site1-S1# show vlan brief (**This is for Site1-S1 and check it on all switches**)

2. Configure Trunking

2.1 Manually configure trunk interface F0/1-2 on Site1-S1 and S2, and F0/3-4 on Site1-S2 and F0/1-2 on S4, and Interface G0/1-2 on Site2-S1 & S2 Switches with EtherChannel (This is for Site1-S1)

- a. Site1-S1(config)# interface range f0/1-2
- b. Site1-S1(config-if-range)# channel-group 1 mode active
Site1-S1(config-if-range)# exit
- c. Site1-S1(config)# interface port-channel 1
- d. Site1-S1(config-if)# switchport mode trunk

- e. Site1-S1(config-if)# switchport mode trunk native vlan 1000
- f. Site1-S1(config-if)# switchport trunk allowed vlan 20,30,40,1000
- g. Site1-S1# show interfaces trunk

2.2 Manually configure Site1-S1 trunk interfaces F0/3, and F0/5. Site1-S4 trunk interface F0/3. Site2-S1 trunk interfaces F0/22 (This configuration is for Site1-S1).

- a. Site1-S1(config)# interface range f0/3, f0/5
Site1-S1(config-if-range)# switchport mode trunk
Site1-S1(config-if-range)# switchport trunk native vlan 1000
Site1-S1(config-if-range)# switchport trunk allowed vlan 20,30,40,1000
- b. Site1-S1# copy running-config startup-config
- c. Site1-S1# show interfaces trunk

3. PortFast and BPDU Guard Configuration on Switches (Do not configure on Site1-S1&S4 interface F0/3) - (This is for Site1-S2)

- a. Site1-S2(config)# spanning-tree portfast bpduguard default

4. Configure Inter-Vlan Routing on Routers Site1-R1 and Site2-R1 (This is for Site1-R1)

- a. Site1-R1(config)# interface g0/0/1
Site1-R1 (config-if)# no shutdown
- b. Site1-R1(config)# interface g0/0/1.20
Site1-R1(config-subif)# description Management Network
Site1-R1(config-subif)# encapsulation dot1q 20
Site1-R1(config-subif)# ip address 192.168.3.1 255.255.255.240
Site1-R1(config-subif)# exit
Site1-R1(config-subif)# interface g0/0/1.30
Site1-R1(config-subif)# encapsulation dot1q 30
Site1-R1(config-subif)# description Operations Network
Site1-R1(config-subif)# ip address 192.168.2.1 255.255.255.192
Site1-R1(config-subif)# exit
Site1-R1(config-subif)# interface g0/0/1.40
Site1-R1(config-subif)# encapsulation dot1q 40
Site1-R1(config-subif)# description Marketing Network
Site1-R1(config-subif)# ip address 192.168.1.1 255.255.255.0
Site1-R1(config-subif)# exit
Site1-R1(config-subif)# interface g0/0/1.1000
Site1-R1(config-subif)# encapsulation dot1q 1000 Native
Site1-R1(config-subif)# description Native Vlan
- c. Site1-R1# show ip interface brief

5. Configure Default Routing

- a. Site1-R1(config)# interface g0/0/0
Site1-R1(config-if)# ip address 192.168.5.253 255.255.255.0
Site1-R1(config-if)# no shutdown
Site1-R1(config-if)# exit

Site1-R1(config)# interface s0/1/0
Site1-R1(config-if)# ip address 192.168.7.254 255.255.255.252
Site1-R1(config-if)# no shutdown
Site1-R1(config-if)# exit

S1Annex-R2(config)# interface s0/1/0
S1Annex-R2(config-if)# ip address 192.168.7.253 255.255.255.252
S1Annex-R2(config-if)# no shutdown
S1Annex-R2(config-if)# exit

Site2-R1 (config)# interface g0/0/1
Site2-R1(config-if)# ip address 192.168.11.1 255.255.255.0
Site2-R1(config-if)# no shutdown
Site2-R1(config-if)# exit
- b. Site1-R1(config)# ip route 0.0.0.0 0.0.0.0 g0/0/0 192.168.5.254
Site1-R1(config)# ip route 192.168.6.0 255.255.255.224 192.168.7.253
S1Annex-R2(config)# ip route 0.0.0.0 0.0.0.0 s0/1/0 192.168.7.254
Site2-R1(config)# ip route 0.0.0.0 0.0.0.0 g0/0/1 192.168.11.2

6. Configure DHCP on Routers (This is for Site1-R1)

- a. Site1-R1(config)#ip dhcp excluded-address 192.168.3.1 192.168.3.2
- b. Site1-R1(config)#ip dhcp pool Management
Site1-R1(config-dhcp)# network 192.168.3.0 255.255.255.240
- c. Site1-R1(config-dhcp)# dns-server 8.8.8.8
- d. Site1-R1(config-dhcp)# default-router 192.168.3.1
Site1-R1(config-dhcp)# domain-name hsk-spares.com
- e. Site1-R1#show ip dhcp binding

Same step to create other configurations for Vlan 30 and Vlan 40

```
Site1-R1(config)#ip dhcp excluded-address 192.168.2.1 192.168.2.2
Site1-R1(config)#ip dhcp pool Operations
Site1-R1(config-dhcp)# network 192.168.2.0 255.255.255.192
Site1-R1(config-dhcp)# dns-server 8.8.8.8
Site1-R1(config-dhcp)# default-router 192.168.2.1
```

```
Site1-R1(config-dhcp)# domain-name hsk-spare.com
Site1-R1(config-dhcp)# end
Site1-R1#show ip dhcp binding

Site1-R1(config)#ip dhcp excluded-address 192.168.1.1
Site1-R1(config)#ip dhcp pool Marketing
Site1-R1(config-dhcp)# network 192.168.1.0 255.255.255.0
Site1-R1(config-dhcp)# dns-server 8.8.8.8
Site1-R1(config-dhcp)# default-router 192.168.1.1
Site1-R1(config-dhcp)# domain-name hsk-spare.com
Site1-R1(config-dhcp)# end
Site1-R1#show ip dhcp binding
```

7. Configure PAT on Routers Site1-R1 and Site2-R1 (This is for Site1-R1)

- a. Site1-R1(config)# access-list 1 permit 192.168.1.0 0.0.0.255
Site1-R1(config)# access-list 1 permit 192.168.2.0 0.0.0.63
Site1-R1(config)# access-list 1 permit 192.168.3.0 0.0.0.15
Site1-R1(config)# access-list 1 permit 192.168.6.0 0.0.0.63
- b. Site1-R1(config)# interface g0/0/0
Site1-R1(config-if)# ip nat outside
- c. Site1-R1(config)# interface g0/0/1.20
Site1-R1(config-if)# ip nat inside

Site1-R1(config)# interface g0/0/1.30
Site1-R1(config-if)# ip nat inside

Site1-R1(config)# interface g0/0/1.40
Site1-R1(config-if)# ip nat inside

Site1-R1(config)# interface g0/0/1.1000
Site1-R1(config-if)# ip nat inside
- d. Site1-R1(config)# ip nat inside source list 1 interface g0/0/0 overload

8. Configure all network devices for basic SSH support (This is for Site1-R1)

- a. Site1-R1(config)# username SSHadmin secret cisco12345
- b. Site1-R1(config)# ip domain name hsk-spare.com
- c. Site1-R1(config)# crypto key generate rsa general-keys modulus 2048
- d. Site1-R1(config)# line vty 0 4
Site1-R1(config-line)# transport input ssh
Site1-R1(config-line)# login local
Site1-R1(config-line)# exit

9. Configure ASA interfaces

- a. NETSEC-ASA(config)# interface GigabitEthernet1/1
NETSEC-ASA(config-if)# nameif OUTSIDE
NETSEC-ASA(config-if)# security-level 0
NETSEC-ASA(config-if)# ip address 172.24.104.4 255.255.255.248
NETSEC-ASA(config-if)# no shutdown

ASA1(config)# interface GigabitEthernet1/3
ASA1(config-if)# nameif OUTSIDE
ASA1(config-if)# security-level 0
ASA1(config-if)# ip address 172.24.160.4 255.255.255.248
ASA1(config-if)# no shutdown
- b. NETSEC-ASA(config)# interface GigabitEthernet1/2
NETSEC-ASA(config-if)# nameif INSIDE
NETSEC-ASA(config-if)# security-level 100
NETSEC-ASA(config-if)# ip address 192.168.5.254 255.255.255.0
NETSEC-ASA(config-if)# no shutdown

ASA1(config)# interface GigabitEthernet1/1
ASA1(config-if)# nameif INSIDE
ASA1(config-if)# security-level 100
ASA1(config-if)# ip address 192.168.11.2 255.255.255.0
ASA1(config-if)# no shutdown
- c. NETSEC-ASA(config)# interface GigabitEthernet1/3
NETSEC-ASA(config-if)# nameif DMZ
NETSEC-ASA(config-if)# security-level 70
NETSEC-ASA(config-if)# ip address 192.168.4.1 255.255.255.192
NETSEC-ASA(config-if)# no shutdown

10. Configure Default route for the ASA

- a. NETSEC-ASA(config)# route OUTSIDE 0.0.0.0 0.0.0.0 172.24.104.1
NETSEC-ASA(config)# route INSIDE 192.168.0.0 255.255.0.0 17192.168.5.253
ASA1(config)# route OUTSIDE 0.0.0.0 0.0.0.0 172.24.160.1
ASA1(config)# route INSIDE 192.168.0.0 255.255.0.0 17192.168.5.253
- b. ASA1(config)# Route INSIDE 192.168.0.0 255.255.0.0 192.168.11.1

11. Configure address translation using PAT and network objects

- a. NETSEC-ASA(config)# object network INSIDE-NET
NETSEC-ASA(config-object)# subnet 192.168.0.0 255.255.0.0
NETSEC-ASA(config-object)# nat (INSIDE, OUTSIDE) dynamic interface
NETSEC-ASA(config-object)# end

```
ASA1(config)# object network INSIDE-NET
ASA1(config-object)# subnet 192.168.0.0 255.255.0.0
ASA1(config-object)# nat (INSIDE, OUTSIDE) dynamic interface
ASA1(config-object)# end
```

- b. NETSEC-ASA(config)# object network DMZ-NET
NETSEC-ASA(config-object)# subnet 192.168.4.0 255.255.255.192
NETSEC-ASA(config-object)# nat (DMZ, OUTSIDE) dynamic interface
NETSEC-ASA(config-object)# end

12. Enable ICMP Inspection in Global Policy

- a. NETSEC-ASA(config)# policy-map global_policy
NETSEC-ASA(config)# class inspection_default
NETSEC-ASA(config)# inspect icmp

```
ASA1(config)# policy-map global_policy
ASA1(config)# class inspection_default
ASA1(config)# inspect icmp
```

```
NETSEC-ASA(config)# show run policy-map
```

13. Configure Static NAT to the DMZ Servers using a network object

- a. NETSEC-ASA(config)# object network INTERNET-SERVER
NETSEC-ASA(config-object)# host 192.168.4.46
NETSEC-ASA(config-object)# nat (DMZ, OUTSIDE) static 172.24.104.6
NETSEC-ASA(config-object)# end

14. Configure an ACL to allow access to the Web servers from the Internet

- a. NETSEC-ASA(config)# access-list OUTSIDE-DMZ extended permit ip any host 192.168.4.46
NETSEC-ASA(config)# access-group OUTSIDE-DMZ in interface OUTSIDE
- b. NETSEC-ASA(config)# access-list DMZ-INSIDE extended permit icmp any any
NETSEC-ASA(config)# access-list DMZ-INSIDE extended permit tcp any any eq 80
NETSEC-ASA(config)# access-list DMZ-INSIDE extended permit tcp any any eq 443
NETSEC-ASA(config)# access-group DMZ-INSIDE in interface DMZ

15. Configure AAA to use local database for authentication and SSH remote access to ASA

- a. NETSEC-ASA(config)# username SSHadmin password cisco12345 (for ASA1 username is admin01)
- b. NETSEC-ASA(config)# aaa authentication ssh console LOCAL
- c. NETSEC-ASA(config)# crypto key generate rsa modulus 2048
NETSEC-ASA(config)# write mem
- e. NETSEC-ASA(config)# ssh 192.168.5.0 255.255.255.0 INSIDE
- f. NETSEC-ASA(config)# ssh timeout 10

16. Configure Site-to-Site VPN

- a. NETSEC-ASA(config)# object-group network SITE1-NET
NETSEC-ASA(config-network-object)# network-object 192.168.1.0 255.255.255.0
NETSEC-ASA(config-network-object)# network-object 192.168.2.0 255.255.255.192
NETSEC-ASA(config-network-object)# network-object 192.168.3.0 255.255.255.240
NETSEC-ASA(config-network-object)# network-object 192.168.4.0 255.255.255.192
NETSEC-ASA(config-network-object)# network-object 192.168.5.0 255.255.255.0
NETSEC-ASA(config-network-object)# network-object 192.168.6.0 255.255.255.192

```
NETSEC-ASA(config)# object-group network SITE2-NET
NETSEC-ASA(config-network-object)# network-object 192.168.10.0 255.255.255.0
NETSEC-ASA(config-network-object)# network-object 192.168.11.0 255.255.255.0
NETSEC-ASA(config-network-object)# network-object 192.168.50.0 255.255.255.0
```

```
ASA1(config)# object-group network SITE2-NET
ASA1(config-network-object)# network-object 192.168.10.0 255.255.255.0
ASA1(config-network-object)# network-object 192.168.11.0 255.255.255.0
ASA1(config-network-object)# network-object 192.168.50.0 255.255.255.0
```

```
ASA1(config)# object-group network SITE1-NET
ASA1(config-network-object)# network-object 192.168.1.0 255.255.255.0
ASA1(config-network-object)# network-object 192.168.2.0 255.255.255.192
ASA1(config-network-object)# network-object 192.168.3.0 255.255.255.240
ASA1(config-network-object)# network-object 192.168.4.0 255.255.255.192
ASA1(config-network-object)# network-object 192.168.5.0 255.255.255.0
ASA1(config-network-object)# network-object 192.168.6.0 255.255.255.192
```

- b. NETSEC-ASA(config)# crypto ikev1 policy 10
NETSEC-ASA(config-isakmp)# authentication pre-share
NETSEC-ASA(config-isakmp)# encryption aes-256
NETSEC-ASA(config-isakmp)# hash sha
NETSEC-ASA(config-isakmp)# group 2
NETSEC-ASA(config-isakmp)# lifetime 86400
NETSEC-ASA(config-isakmp)# exit

ASA1(config)# crypto ikev1 policy 10
ASA1(config-isakmp)# authentication pre-share
ASA1(config-isakmp)# encryption aes-256
ASA1(config-isakmp)# hash sha
ASA1(config-isakmp)# group 2
ASA1(config-isakmp)# lifetime 86400
ASA1(config-isakmp)# exit
- c. NETSEC-ASA(config)# crypto ikev1 enable OUTSIDE
ASA1(config)# crypto ikev1 enable OUTSIDE
- d. NETSEC-ASA(config)# crypto ipsec ikev1 transform-set VPN-SET esp-aes-256 esp-sha-hmac
ASA1(config)# crypto ipsec ikev1 transform-set VPN-SET esp-aes-256 esp-sha-hmac
- e. NETSEC-ASA(config)# access-list VPN-TRAFFIC extended permit ip object-group SITE1-NET
object-group SITE2-NET
NETSEC-ASA(config)# access-list INSIDE-IN extended permit ip object-group SITE1-NET
object-group SITE2-NET
NETSEC-ASA(config)# access-list INSIDE-IN extended permit ip any any
NETSEC-ASA(config)# access-list OUTSIDE-DMZ extended permit udp any any eq 500
NETSEC-ASA(config)# access-list OUTSIDE-DMZ extended permit udp any any eq 4500
NETSEC-ASA(config)# access-list OUTSIDE-DMZ extended permit esp any any
NETSEC-ASA(config)# access-list DMZ-INSIDE extended permit ip 192.168.4.0
255.255.255.192 object-group SITE2-NET
NETSEC-ASA(config)# access-group INSIDE-IN in interface INSIDE

ASA1(config)# access-list VPN-TRAFFIC extended permit ip object-group SITE2-NET object-
group SITE1-NET
ASA1(config)# access-list INSIDE-IN extended permit ip object-group SITE2-NET object-group
SITE1-NET
ASA1(config)# access-list INSIDE-IN extended permit ip any any
ASA1(config)# access-list OUTSIDE-IN extended permit udp any any eq 500
ASA1(config)# access-list OUTSIDE-IN extended permit udp any any eq 4500
ASA1(config)# access-list OUTSIDE-IN extended permit udp any any eq 4500
ASA1(config)# access-group INSIDE-IN in interface INSIDE
ASA1(config)# access-group OUTSIDE-IN in interface OUTSIDE

- f. NETSEC-ASA(config)# tunnel-group 172.24.160.4 type ipsec-l2l
NETSEC-ASA(config)# tunnel-group 172.24.160.4 ipsec-attributes
NETSEC-ASA(ipsec-attributes)# ikev1 pre-shared-key cisco123

ASA1(config)# tunnel-group 172.24.104.4 type ipsec-l2l
ASA1(config)# tunnel-group 172.24.104.4 ipsec-attributes
ASA1(ipsec-attributes)# ikev1 pre-shared-key cisco123
- g. NETSEC-ASA(config)# crypto map VPN-MAP 10 match address VPN-TRAFFIC
NETSEC-ASA(config)# crypto map VPN-MAP 10 set peer 172.24.160.4
NETSEC-ASA(config)# crypto map VPN-MAP 10 set ikev1 transform-set VPN-SET
NETSEC-ASA(config)# crypto map VPN-MAP interface OUTSIDE

ASA1(config)# crypto map VPN-MAP 10 match address VPN-TRAFFIC
ASA1(config)# crypto map VPN-MAP 10 set peer 172.24.104.4
ASA1(config)# crypto map VPN-MAP 10 set ikev1 transform-set VPN-SET
ASA1(config)# crypto map VPN-MAP interface OUTSIDE
- h. NETSEC-ASA(config)# nat (INSIDE,OUTSIDE) source static SITE1-NET SITE1-NET
destination static SITE2-NET SITE2-NET no-proxy-arp
NETSEC-ASA(config)# nat (DMZ,OUTSIDE) source static SITE1-NET SITE1-NET destination
static SITE2-NET SITE2-NET no-proxy-arp

ASA1(config)# nat (INSIDE,OUTSIDE) source static SITE2-NET SITE2-NET destination static
SITE1-NET SITE1-NET no-proxy-arp
- i. Site1-R1(config)# ip route 192.168.10.0 255.255.255.0 192.168.5.254
Site1-R1(config)# ip route 192.168.11.0 255.255.255.0 192.168.5.254
Site1-R1(config)# ip route 192.168.50.0 255.255.255.0 192.168.5.254
Site1-R1(config)# ip route 192.168.6.0 255.255.255.0 192.168.7.253

Site2-R1(config)# ip route 192.168.1.0 255.255.255.0 192.168.11.2
Site2-R1(config)# ip route 192.168.2.0 255.255.255.192 192.168.11.2
Site2-R1(config)# ip route 192.168.3.0 255.255.255.240 192.168.11.2
Site2-R1(config)# ip route 192.168.4.0 255.255.255.192 192.168.11.2
Site2-R1(config)# ip route 192.168.5.0 255.255.255.0 192.168.11.2
Site2-R1(config)# ip route 192.168.6.0 255.255.255.192 192.168.11.2

17. Configure OSPF on Routers

- a. Site1-R1(config)# router ospf 101
Site2-R1(config-router)# router ospf 101
S1Annex-R2(config-router)# router ospf 101
- b. Site1-R1(config-router)# router id 2.2.2.2

```
Site2-R1(config-router)# router id 1.1.1.1
S1Annex-R2(config-router)# router id 3.3.3.3
```

- c. Site1-R1(config-router)# network 192.168.1.0 0.0.0.255 area 0
Site1-R1(config-router)# network 192.168.2.0 0.0.0.63 area 0
Site1-R1(config-router)# network 192.168.3.0 0.0.0.15 area 0
Site1-R1(config-router)# network 192.168.5.252 0.0.0.3 area 0
Site1-R1(config-router)# network 192.168.7.252 0.0.0.3 area 0

```
Site2-R1(config-router)# network 192.168.10.0 0.0.0.255 area 0
Site2-R1(config-router)# network 192.168.20.0 0.0.0.63 area 0
Site2-R1(config-router)# network 192.168.30.0 0.0.0.15 area 0
Site2-R1(config-router)# network 172.24.104.0 0.0.0.7 area 0
```

```
S1Annex-R2(config-router)# network 192.168.6.0 0.0.0.31 area 0
S1Annex-R2(config-router)# network 192.168.7.252 0.0.0.3 area 0
```

18. Configure Business Partners to access the B2B web.

- a. NETSEC-ASA(config)# object network B2B-SERVER
NETSEC-ASA(config-object)# host 192.168.4.47
NETSEC-ASA(config-object)# nat (DMZ, OUTSIDE) static 172.24.104.5
NETSEC-ASA(config-object)# end
- b. NETSEC-ASA(config)# access-list OUTSIDE-DMZ extended permit ip host 172.24.56.4 host 192.168.4.47
NETSEC-ASA(config)# access-list OUTSIDE-DMZ extended permit ip host 172.24.176.4 host 192.168.4.47

NETSEC-ASA(config)# access-group OUTSIDE-DMZ in interface OUTSIDE

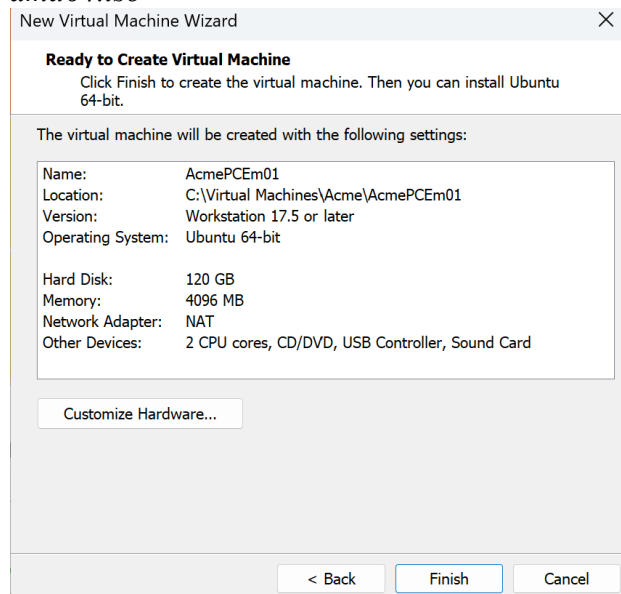
19. Configure NTP in the Network

- ```
Site1-R1# clock set 00:00:00 Month Date Year
Site1-R1# configuration terminal
Site1-R1# (config)# ntp master 4
```
- a. NETSEC-ASA(config)# ntp server 192.168.5.253 source INSIDE  
ASA1(config)# ntp server 192.168.5.253 source INSIDE
  - b. S1Annex-R2(config)# ntp server 192.168.5.253 prefer  
Site2-R1(config)# ntp server 192.168.5.253 prefer  
Site1-S1(config)# ntp server 192.168.5.253 prefer  
Site1-S2(config)# ntp server 192.168.5.253 prefer

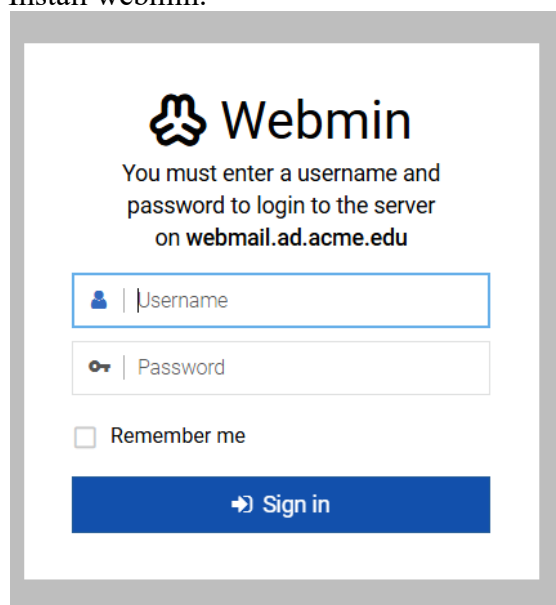
```
Site1-S3(config)# ntp server 192.168.5.253 prefer
Site1-S4(config)# ntp server 192.168.5.253 prefer
S1Annex-S5(config)# ntp server 192.168.5.253 prefer
Site2-S1(config)# ntp server 192.168.5.253 prefer
Site2-S2(config)# ntp server 192.168.5.253 prefer
```

## 20. Configure the Web server

- Download and install *VMware-player-full-17.5.0-22583795.exe*
- The details of the virtual machine are as follows, download and install *ubuntu-24.04.2-live-server-amd64.iso*



- Install webmin.



- d. Create new virtual interfaces with eth0 – eth3 and details of the interfaces as follows.

| Device          | Interface         | IP Address   | Subnet Mask     | Broadcast address |
|-----------------|-------------------|--------------|-----------------|-------------------|
| Virtual Machine | ens33 -Ethernet   | 192.168.4.3  | 255.255.255.192 | 192.168.4.63      |
|                 | ens33.0 -Intranet | 192.168.4.45 | 255.255.255.192 | 192.168.4.63      |
|                 | ens33.1 -Internet | 192.168.4.46 | 255.255.255.192 | 192.168.4.63      |
|                 | ens33.2 -B2B      | 192.168.4.47 | 255.255.255.192 | 192.168.4.63      |

## Appendix – 3

### Each Device Configuration

#### **Site1-R1**

```
router(config)# hostname Site1-R1
Site1-R1(config)# no ip domain lookup
Site1-R1(config)# enable secret class
Site1-R1(config)# line console 0
Site1-R1(config-line)# password cisco
Site1-R1(config-line)# login
Site1-R1(config)# line vty 0 4
Site1-R1(config-line)# password cisco
Site1-R1(config-line)# login
Site1-R1(config)# service password-encryption
Site1-R1(config)# banner motd $ Unauthorized access is Prohibited! $
```

```
Site1-R1(config)# interface g0/0/1
Site1-R1(config-if)# no shutdown
```

```
Site1-R1(config)# interface g0/0/1.20
Site1-R1(config-subif)# description Management Network
Site1-R1(config-subif)# encapsulation dot1q 20
Site1-R1(config-subif)# ip address 192.168.3.1 255.255.255.240
Site1-R1(config-subif)# exit
Site1-R1(config-subif)# interface g0/0/1.30
Site1-R1(config-subif)# encapsulation dot1q 30
Site1-R1(config-subif)# description Operations Network
Site1-R1(config-subif)# ip address 192.168.2.1 255.255.255.192
Site1-R1(config-subif)# exit
Site1-R1(config-subif)# interface g0/0/1.40
Site1-R1(config-subif)# encapsulation dot1q 40
Site1-R1(config-subif)# description Marketing Network
Site1-R1(config-subif)# ip address 192.168.1.1 255.255.255.0
Site1-R1(config-subif)# exit
Site1-R1(config-subif)# interface g0/0/1.1000
Site1-R1(config-subif)# encapsulation dot1q 1000 Native
Site1-R1(config-subif)# description Native Vlan
Site1-R1(config-subif)# exit
```

```
Site1-R1(config)# interface g0/0/0
Site1-R1(config-if)# ip address 192.168.5.253 255.255.255.0
Site1-R1(config-if)# no shutdown
Site1-R1(config-if)# exit
```

```
Site1-R1(config)# interface S0/1/0
Site1-R1(config-if)# ip address 192.168.7.254 255.255.255.252
Site1-R1(config-if)# no shutdown
Site1-R1(config-if)# exit
```

```
Site1-R1(config)# access-list 1 permit 192.168.1.0 0.0.0.255
Site1-R1(config)# access-list 1 permit 192.168.2.0 0.0.0.63
Site1-R1(config)# access-list 1 permit 192.168.3.0 0.0.0.15
Site1-R1(config)# access-list 1 permit 192.168.6.0 0.0.0.63
```

```
Site1-R1(config)# interface g0/0/0
Site1-R1(config-if)# ip nat outside
```

```
Site1-R1(config)# interface g0/0/1.20
Site1-R1(config-if)# ip nat inside
```

```
Site1-R1(config)# interface g0/0/1.30
Site1-R1(config-if)# ip nat inside
```

```
Site1-R1(config)# interface g0/0/1.40
Site1-R1(config-if)# ip nat inside
```

```
Site1-R1(config)# interface g0/0/1.1000
Site1-R1(config-if)# ip nat inside
```

```
Site1-R1(config)# ip nat inside source list 1 interface g0/0/0 overload
```

```
Site1-R1(config)# ip route 0.0.0.0 0.0.0.0 g0/0/0 192.168.5.254
Site1-R1(config)# ip route 192.168.6.0 255.255.255.224 192.168.7.253
Site1-R1(config)# ip route 192.168.10.0 255.255.255.0 192.168.5.254
Site1-R1(config)# ip route 192.168.11.0 255.255.255.0 192.168.5.254
Site1-R1(config)# ip route 192.168.50.0 255.255.255.0 192.168.5.254
```

```
Site1-R1(config)#ip dhcp excluded-address 192.168.3.1 192.168.3.2
Site1-R1(config)#ip dhcp pool Management
Site1-R1(config-dhcp)# network 192.168.3.0 255.255.255.240
Site1-R1(config-dhcp)# dns-server 8.8.8.8
Site1-R1(config-dhcp)# default-router 192.168.3.1
Site1-R1(config-dhcp)# domain-name hsk-spare.com
Site1-R1#show ip dhcp binding
```

```
Site1-R1(config)#ip dhcp excluded-address 192.168.2.1 192.168.2.2
Site1-R1(config)#ip dhcp pool Operations
Site1-R1(config-dhcp)# network 192.168.2.0 255.255.255.192
Site1-R1(config-dhcp)# dns-server 8.8.8.8
```

```
Site1-R1(config-dhcp)# default-router 192.168.2.1
Site1-R1(config-dhcp)# domain-name hsk-spare.com
Site1-R1(config-dhcp)# end
```

```
Site1-R1(config)#ip dhcp excluded-address 192.168.1.1
Site1-R1(config)#ip dhcp pool Marketing
Site1-R1(config-dhcp)# network 192.168.1.0 255.255.255.0
Site1-R1(config-dhcp)# dns-server 8.8.8.8
Site1-R1(config-dhcp)# default-router 192.168.1.1
Site1-R1(config-dhcp)# domain-name hsk-spare.com
```

```
Site1-R1(config)# username SSHadmin secret cisco12345
Site1-R1(config)# ip domain name hsk-spare.com
Site1-R1(config)# crypto key generate rsa general-keys modulus 2048
Site1-R1(config)# line vty 0 4
Site1-R1(config-line)# transport input ssh
Site1-R1(config-line)# login local
Site1-R1(config-line)# exit
```

```
Site1-R1(config)# router ospf 101
Site1-R1(config-router)# router id 2.2.2.2
Site1-R1(config-router)# network 192.168.1.0 0.0.0.255 area 0
Site1-R1(config-router)# network 192.168.2.0 0.0.0.63 area 0
Site1-R1(config-router)# network 192.168.3.0 0.0.0.15 area 0
Site1-R1(config-router)# network 192.168.5.252 0.0.0.3 area 0
Site1-R1(config-router)# network 192.168.7.252 0.0.0.3 area 0
Site1-R1(config-router)# end
```

```
Site1-R1# clock set 00:00:00 Month Date Year
Site1-R1# configuration terminal
Site1-R1# (config)# ntp master 4
Site1-R1# (config)# exit
```

```
Site1-R1# copy running-config startup-config
```

### **S1Annex-R2**

```
router(config)# hostname S1Annex-R2
S1Annex-R2(config)# no ip domain lookup
S1Annex-R2 (config)# enable secret class
S1Annex-R2(config)# line console 0
S1Annex-R2(config-line)# password cisco
S1Annex-R2(config-line)# login
S1Annex-R2(config)# line vty 0 4
S1Annex-R2(config-line)# password cisco
S1Annex-R2(config-line)# login
S1Annex-R2(config)# service password-encryption
S1Annex-R2(config)# banner motd $ Unauthorized access is Prohibited! $
```

```
S1Annex-R2(config)# interface s0/1/0
S1Annex-R2(config-if)# ip address 192.168.7.253 255.255.255.252
S1Annex-R2(config-if)# no shutdown
S1Annex-R2(config-if)# exit
```

```
S1Annex-R2(config)# ip route 0.0.0.0 0.0.0.0 s0/1/0 192.168.7.254
```

```
S1Annex-R2(config)#ip dhcp excluded-address 192.168.6.1 192.168.6.2
S1Annex-R2(config)#ip dhcp pool R&D
S1Annex-R2(config-dhcp)# network 192.168.6.0 255.255.255.224
S1Annex-R2(config-dhcp)# dns-server 8.8.8.8
S1Annex-R2(config-dhcp)# default-router 192.168.6.1
S1Annex-R2(config-dhcp)# domain-name hsk-spare.com
```

```
S1Annex-R2(config)# username SSHadmin secret cisco12345
S1Annex-R2(config)# ip domain name hsk-spare.com
S1Annex-R2(config)# crypto key generate rsa general-keys modulus 2048
S1Annex-R2(config)# line vty 0 4
S1Annex-R2(config-line)# transport input ssh
S1Annex-R2(config-line)# login local
S1Annex-R2(config-line)# exit
```

```
S1Annex-R2(config-router)# router ospf 101
S1Annex-R2(config-router)# router id 3.3.3.3
S1Annex-R2(config-router)# network 192.168.6.0 0.0.0.31 area 0
S1Annex-R2(config-router)# network 192.168.7.252 0.0.0.3 area 0
S1Annex-R2(config-router)# exit
```

```
S1Annex-R2(config)# ntp server 192.168.5.253 prefer
S1Annex-R2(config)# exit
```

```
S1Annex-R2# copy running-config startup-config
```

### Site2-R1

```
router(config)# hostname Site2-R1
Site2-R1(config)# no ip domain lookup
Site2-R1(config)# Enable algorithm-type scrypt secret class12345
Site2-R1(config)# line console 0
Site2-R1(config-line)# password cisco
Site2-R1(config-line)# login
Site2-R1(config)# line vty 0 4
Site2-R1(config-line)# password cisco
Site2-R1(config-line)# login
Site2-R1(config)# service password-encryption
Site2-R1(config)# banner motd $ Unauthorized access is Prohibited! $
```

```
Site2-R1(config)# interface g0/0/0
Site2-R1 (config-if)# no shutdown
```

```
Site2-R1(config)# interface g0/0/0.10
Site2-R1(config-subif)# description Management Network
Site2-R1(config-subif)# encapsulation dot1q 10
Site2-R1(config-subif)# ip address 192.168.10.1 255.255.255.0
Site2-R1(config-subif)# exit
Site2-R1(config-subif)# interface g0/0/0.50
Site2-R1(config-subif)# encapsulation dot1q 50
Site2-R1(config-subif)# description Business Network
Site2-R1(config-subif)# ip address 192.168.50.1 255.255.255.0
Site2-R1(config-subif)# exit
Site2-R1(config-subif)# interface g0/0/0.2
Site2-R1(config-subif)# encapsulation dot1q 2 Native
Site2-R1(config-subif)# description Native Vlan
Site2-R1(config-subif)# exit
```

```
Site2-R1 (config)# interface g0/0/1
Site2-R1(config-if)# ip address 192.168.11.1 255.255.255.0
Site2-R1(config-if)# no shutdown
Site2-R1(config-if)# exit
```

```
Site2-R1(config)# ip route 0.0.0.0 0.0.0.0 g0/0/1 192.168.11.2
Site2-R1(config)# ip route 192.168.1.0 255.255.255.0 192.168.11.2
```

```
Site2-R1(config)# ip route 192.168.2.0 255.255.255.192 192.168.11.2
Site2-R1(config)# ip route 192.168.3.0 255.255.255.240 192.168.11.2
Site2-R1(config)# ip route 192.168.4.0 255.255.255.192 192.168.11.2
Site2-R1(config)# ip route 192.168.5.0 255.255.255.0 192.168.11.2
Site2-R1(config)# ip route 192.168.6.0 255.255.255.192 192.168.11.2
```

```
Site2-R1(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.2 192.168.10.3
Site2-R1(config)#ip dhcp pool IT-MAN-10
Site2-R1(config-dhcp)# network 192.168.10.0 255.255.255.0
Site2-R1(config-dhcp)# dns-server 8.8.8.8
Site2-R1(config-dhcp)# default-router 192.168.10.1
Site2-R1(config-dhcp)# domain-name example.com
Site2-R1#show ip dhcp binding
```

```
Site2-R1(config)#ip dhcp excluded-address 192.168.50.1
Site2-R1(config)#ip dhcp pool BUSS-MAN-50
Site2-R1(config-dhcp)# network 192.168.50.0 255.255.255.0
Site2-R1(config-dhcp)# dns-server 8.8.8.8
Site2-R1(config-dhcp)# default-router 192.168.50.1
Site2-R1(config-dhcp)# domain-name example.com
Site2-R1(config-dhcp)# exit
```

```
Site2-R1(config)# username admin01 secret cisco12345
Site2-R1(config)# ip domain name example.com
Site2-R1(config)# crypto key generate rsa general-keys modulus 1024
Site2-R1(config)# line vty 0 4
Site2-R1(config-line)# transport input ssh
Site2-R1(config-line)# login local
Site2-R1(config-line)# line console 0
Site2-R1(config-line)# login local
Site2-R1(config-line)# exec-timeout 5 0
Site2-R1(config-line)# logging synchronous
Site2-R1(config-line)# exit
```

```
Site2-R1(config-router)# router ospf 101
Site2-R1(config-router)# router id 2.2.2.2
Site2-R1(config-router)# network 192.168.10.0 0.0.0.255 area 0
Site2-R1(config-router)# network 192.168.20.0 0.0.0.63 area 0
Site2-R1(config-router)# network 192.168.30.0 0.0.0.15 area 0
Site2-R1(config-router)# network 172.24.104.0 0.0.0.7 area 0
Site2-R1(config-router)# exit
```

```
Site2-R1(config)# ntp server 192.168.5.253 prefer
Site2-R1(config)# exit
```

```
Site1-R1# copy running-config startup-config
```

### **NETSEC-ASA (Site1)**

```
NETSEC-ASA(config)# hostname NETSEC-ASA
NETSEC-ASA(config)# domain-name hsk-spare.com
NETSEC-ASA(config)# enable password cisco
NETSEC-ASA(config)# service password-encryption
NETSEC-ASA(config)# banner motd $ Unauthorized access is Prohibited! $
```

```
NETSEC-ASA(config)# interface GigabitEthernet1/1
NETSEC-ASA(config-if)# nameif OUTSIDE
NETSEC-ASA(config-if)# security-level 0
NETSEC-ASA(config-if)# ip address 172.24.104.4 255.255.255.248
NETSEC-ASA(config-if)# no shutdown
NETSEC-ASA(config)# interface GigabitEthernet1/2
NETSEC-ASA(config-if)# nameif INSIDE
NETSEC-ASA(config-if)# security-level 100
NETSEC-ASA(config-if)# ip address 192.168.5.254 255.255.255.0
NETSEC-ASA(config-if)# no shutdown
NETSEC-ASA(config)# interface GigabitEthernet1/3
NETSEC-ASA(config-if)# nameif DMZ
NETSEC-ASA(config-if)# security-level 70
NETSEC-ASA(config-if)# ip address 192.168.4.1 255.255.255.192
NETSEC-ASA(config-if)# no shutdown
```

```
NETSEC-ASA(config)# route OUTSIDE 0.0.0.0 0.0.0.0 172.24.104.1
NETSEC-ASA(config)# route INSIDE 192.168.0.0 255.255.0.0 17192.168.5.253
```

```
NETSEC-ASA(config)# object network INSIDE-NET
NETSEC-ASA(config-object)# subnet 192.168.5.0 255.255.255.0
NETSEC-ASA(config-object)# nat (INSIDE, OUTSIDE) dynamic interface
NETSEC-ASA(config-object)# end
```

```
NETSEC-ASA(config)# object network DMZ-NET
NETSEC-ASA(config)# subnet 192.168.4.0 255.255.255.192
NETSEC-ASA(config)# nat (DMZ,OUTSIDE) dynamic interface
NETSEC-ASA(config)# policy-map global_policy
```

```
NETSEC-ASA(config-pmap)# class inspection_default
NETSEC-ASA(config-pmap-c)# inspect icmp
NETSEC-ASA(config)# show run policy-map
```

```
NETSEC-ASA(config)# object network INTERNET-SERVER
NETSEC-ASA(config-object)# host 192.168.4.46
NETSEC-ASA(config-object)# nat (DMZ, OUTSIDE) static 172.24.104.6
NETSEC-ASA(config-object)# exit
```

```
NETSEC-ASA(config)# access-list OUTSIDE-DMZ permit ip any host 192.168.4.46
NETSEC-ASA(config)# access-group OUTSIDE-DMZ in interface OUTSIDE
```

```
NETSEC-ASA(config)# access-list DMZ-INSIDE extended permit tcp any any eq 80
NETSEC-ASA(config)# access-list DMZ-INSIDE extended permit tcp any any eq 443
NETSEC-ASA(config)# access-list DMZ-INSIDE extended permit icmp any any
NETSEC-ASA(config)# access-group DMZ-INSIDE in interface DMZ
NETSEC-ASA(config)# clear xlate
```

```
NETSEC-ASA(config)# username admin password cisco12345
NETSEC-ASA(config)# aaa authentication ssh console LOCAL
NETSEC-ASA(config)# crypto key generate rsa modulus 2048
NETSEC-ASA(config)# write mem
NETSEC-ASA(config)# ssh 192.168.5.0 255.255.255.0 INSIDE
NETSEC-ASA(config)# ssh timeout 10
```

```
NETSEC-ASA(config)# object-group network SITE1-NET
NETSEC-ASA(config-network-object)# network-object 192.168.1.0 255.255.255.0
NETSEC-ASA(config-network-object)# network-object 192.168.2.0 255.255.255.192
NETSEC-ASA(config-network-object)# network-object 192.168.3.0 255.255.255.240
NETSEC-ASA(config-network-object)# network-object 192.168.4.0 255.255.255.192
NETSEC-ASA(config-network-object)# network-object 192.168.5.0 255.255.255.0
NETSEC-ASA(config-network-object)# network-object 192.168.6.0 255.255.255.192
```

```
NETSEC-ASA(config)# object-group network SITE2-NET
NETSEC-ASA(config-network-object)# network-object 192.168.10.0 255.255.255.0
NETSEC-ASA(config-network-object)# network-object 192.168.11.0 255.255.255.0
NETSEC-ASA(config-network-object)# network-object 192.168.50.0 255.255.255.0
```

```
NETSEC-ASA(config)# crypto ikev1 policy 10
NETSEC-ASA(config-isakmp)# authentication pre-share
NETSEC-ASA(config-isakmp)# encryption aes-256
NETSEC-ASA(config-isakmp)# hash sha
```

```
NETSEC-ASA(config-isakmp)# group 2
NETSEC-ASA(config-isakmp)# lifetime 86400
NETSEC-ASA(config-isakmp)# exit
```

```
NETSEC-ASA(config)# crypto ikev1 enable OUTSIDE
```

```
NETSEC-ASA(config)# crypto ipsec ikev1 transform-set VPN-SET esp-aes-256 esp-sha-hmac
```

```
NETSEC-ASA(config)# access-list VPN-TRAFFIC extended permit ip object-group SITE1-NET object-
group SITE2-NET
NETSEC-ASA(config)# access-list INSIDE-IN extended permit ip object-group SITE1-NET object-group
SITE2-NET
NETSEC-ASA(config)# access-list INSIDE-IN extended permit ip any any
NETSEC-ASA(config)# access-list OUTSIDE-DMZ extended permit udp any any eq 500
NETSEC-ASA(config)# access-list OUTSIDE-DMZ extended permit udp any any eq 4500
NETSEC-ASA(config)# access-list OUTSIDE-DMZ extended permit esp any any
NETSEC-ASA(config)# access-list DMZ-INSIDE extended permit ip 192.168.4.0 255.255.255.192
object-group SITE2-NET
NETSEC-ASA(config)# access-group INSIDE-IN in interface INSIDE
```

```
NETSEC-ASA(config)# tunnel-group 172.24.160.4 type ipsec-l2l
NETSEC-ASA(config)# tunnel-group 172.24.160.4 ipsec-attributes
NETSEC-ASA(ipsec-attributes)# ikev1 pre-shared-key cisco123
```

```
NETSEC-ASA(config)# crypto map VPN-MAP 10 match address VPN-TRAFFIC
NETSEC-ASA(config)# crypto map VPN-MAP 10 set peer 172.24.160.4
NETSEC-ASA(config)# crypto map VPN-MAP 10 set ikev1 transform-set VPN-SET
NETSEC-ASA(config)# crypto map VPN-MAP interface OUTSIDE
```

```
NETSEC-ASA(config)# nat (INSIDE,OUTSIDE) source static SITE1-NET SITE1-NET destination static
SITE2-NET SITE2-NET no-proxy-arp
NETSEC-ASA(config)# nat (DMZ,OUTSIDE) source static SITE1-NET SITE1-NET destination static
SITE2-NET SITE2-NET no-proxy-arp
```

```
NETSEC-ASA(config)# object network B2B-SERVER
NETSEC-ASA(config-object)# host 192.168.4.47
NETSEC-ASA(config-object)# nat (DMZ, OUTSIDE) static 172.24.104.5
NETSEC-ASA(config-object)# end
```

```
NETSEC-ASA(config)# access-list OUTSIDE-DMZ extended permit ip host 172.24.56.4 host
192.168.4.47
NETSEC-ASA(config)# access-list OUTSIDE-DMZ extended permit ip host 172.24.176.4 host
192.168.4.47
NETSEC-ASA(config)# access-group OUTSIDE-DMZ in interface OUTSIDE
```

```
NETSEC-ASA(config)# ntp server 192.168.5.253 source INSIDE
```

```
NETSEC-ASA(config)# write memory
```

### **ASA1 (Site2)**

```
ASA1(config)# hostname ASA1
```

```
ASA1(config)# domain-name hsk-spare.com
```

```
ASA1(config)# enable password class
```

```
ASA1(config)# domain name example.com
```

```
ASA1(config)# service password-encryption
```

```
ASA1(config)# banner motd $ Unauthorized access is Prohibited! $
```

```
ASA1(config)# interface GigabitEthernet1/1
```

```
ASA1(config-if)# nameif INSIDE
```

```
ASA1(config-if)# security-level 100
```

```
ASA1(config-if)# ip address 192.168.11.2 255.255.255.0
```

```
ASA1(config-if)# no shutdown
```

```
ASA1(config)# interface GigabitEthernet1/3
```

```
ASA1(config-if)# nameif OUTSIDE
```

```
ASA1(config-if)# security-level 0
```

```
ASA1(config-if)# ip address 172.24.160.4 255.255.255.248
```

```
ASA1(config-if)# no shutdown
```

```
ASA1(config)# route OUTSIDE 0.0.0.0 0.0.0.0 172.24.160.1
```

```
ASA1(config)# Route INSIDE 192.168.0.0 255.255.0.0 192.168.11.1
```

```
ASA1(config)# route OUTSIDE 0.0.0.0 0.0.0.0 172.24.160.1
```

```
ASA1(config)# route INSIDE 192.168.0.0 255.255.0.0 172.168.5.253
```

```
ASA1(config)# object network INSIDE-NET
```

```
ASA1(config-object)# subnet 192.168.11.0 255.255.255.0
```

```
ASA1(config-object)# nat (INSIDE, OUTSIDE) dynamic interface
```

```
ASA1(config-object)# end
```

```
ASA1(config)# policy-map global_policy
```

```
ASA1(config-pmap)# class inspection_default
```

```
ASA1(config-pmap-c)# inspect icmp
```

```
ASA1(config)# show run policy-map
```

```
ASA1(config)# username admin password cisco12345
ASA1(config)# aaa authentication ssh console LOCAL
ASA1(config)# crypto key generate rsa modulus 2048
ASA1(config)# write mem
ASA1(config)# ssh 192.168.10.0 255.255.255.0 INSIDE
ASA1(config)# ssh timeout 5
```

```
ASA1(config)# object-group network SITE2-NET
ASA1(config-network-object)# network-object 192.168.10.0 255.255.255.0
ASA1(config-network-object)# network-object 192.168.11.0 255.255.255.0
ASA1(config-network-object)# network-object 192.168.50.0 255.255.255.0
```

```
ASA1(config)# object-group network SITE1-NET
ASA1(config-network-object)# network-object 192.168.1.0 255.255.255.0
ASA1(config-network-object)# network-object 192.168.2.0 255.255.255.192
ASA1(config-network-object)# network-object 192.168.3.0 255.255.255.240
ASA1(config-network-object)# network-object 192.168.4.0 255.255.255.192
ASA1(config-network-object)# network-object 192.168.5.0 255.255.255.0
ASA1(config-network-object)# network-object 192.168.6.0 255.255.255.192
```

```
ASA1(config)# crypto ikev1 policy 10
ASA1(config-isakmp)# authentication pre-share
ASA1(config-isakmp)# encryption aes-256
ASA1(config-isakmp)# hash sha
ASA1(config-isakmp)# group 2
ASA1(config-isakmp)# lifetime 86400
ASA1(config-isakmp)# exit
```

```
ASA1(config)# crypto ikev1 enable OUTSIDE
```

```
ASA1(config)# crypto ipsec ikev1 transform-set VPN-SET esp-aes-256 esp-sha-hmac
```

```
ASA1(config)# access-list VPN-TRAFFIC extended permit ip object-group SITE2-NET object-group SITE1-NET
ASA1(config)# access-list INSIDE-IN extended permit ip object-group SITE2-NET object-group SITE1-NET
ASA1(config)# access-list INSIDE-IN extended permit ip any any
ASA1(config)# access-list OUTSIDE-IN extended permit udp any any eq 500
ASA1(config)# access-list OUTSIDE-IN extended permit udp any any eq 4500
ASA1(config)# access-list OUTSIDE-IN extended permit udp any any eq 4500
ASA1(config)# access-group INSIDE-IN in interface INSIDE
ASA1(config)# access-group OUTSIDE-IN in interface OUTSIDE
```

```
ASA1(config)# tunnel-group 172.24.104.4 type ipsec-l2l
ASA1(config)# tunnel-group 172.24.104.4 ipsec-attributes
ASA1(ipsec-attributes)# ikev1 pre-shared-key cisco123
```

```
ASA1(config)# crypto map VPN-MAP 10 match address VPN-TRAFFIC
ASA1(config)# crypto map VPN-MAP 10 set peer 172.24.104.4
ASA1(config)# crypto map VPN-MAP 10 set ikev1 transform-set VPN-SET
ASA1(config)# crypto map VPN-MAP interface OUTSIDE
```

```
ASA1(config)# nat (INSIDE,OUTSIDE) source static SITE2-NET SITE2-NET destination static SITE1-
NET SITE1-NET no-proxy-arp
```

```
ASA1(config)# ntp server 192.168.5.253 source OUTSIDE
```

```
ASA1(config)# write memory
```

### Site1-S1

```
switch(config)# hostname Site1-S1
Site1-S1(config)# no ip domain lookup
Site1-S1(config)# enable secret class
Site1-S1(config)# line console 0
Site1-S1(config-line)# password cisco
Site1-S1(config-line)# login
Site1-S1(config)# line vty 0 15
Site1-S1(config-line)# password cisco
Site1-S1(config-line)# login
Site1-S1(config)# service password-encryption
Site1-S1(config)# banner motd $ Unauthorized access is Prohibited! $
Site1-S1(config)# exit
Site1-S1# copy running-config startup-config
```

```
Site1-S1(config)# vlan 20
Site1-S1(config-vlan)# name Management
Site1-S1(config-vlan)# vlan 30
Site1-S1(config-vlan)# name Operations
Site1-S1(config-vlan)# vlan 40
Site1-S1(config-vlan)# name Marketing
Site1-S1(config-vlan)# vlan 999
Site1-S1(config-vlan)# name ParkingLot
Site1-S1(config-vlan)# vlan 1000
Site1-S1(config-vlan)# name Native
```

```
Site1-S1(config-vlan)# exit
Site1-S1(config)# interface vlan 30
Site1-S1(config-if)# ip address dhcp
Site1-S1(config-if)# no shutdown
Site1-S1(config-if)# exit
Site1-S1(config)# ip default-gateway 192.168.2.1
```

```
Site1-S1(config)# interface range f0/6 - 14
Site1-S1(config-if-range)# switchport mode access
Site1-S1(config-if-range)# switchport access vlan 30
Site1-S1(config-if-range)# exit
Site1-S1(config)# interface range f0/15 - 24
Site1-S1(config-if-range)# switchport mode access
Site1-S1(config-if-range)# switchport access vlan 40
Site1-S1(config-if-range)# exit
Site1-S1(config)# interface range f0/4, g0/1 - 2
Site1-S1(config-if-range)# switchport mode access
Site1-S1(config-if-range)# switchport access vlan 999
Site1-S1(config-if-range)# shutdown
Site1-S1(config-if-range)# exit
```

```
Site1-S1(config)# interface range f0/1-2
Site1-S1(config-if-range)# channel-group 1 mode active
Site1-S1(config-if-range)# exit
Site1-S1(config)# interface port-channel 1
Site1-S1(config-if)# switchport mode trunk
Site1-S1(config-if)# switchport mode trunk native vlan 1000
Site1-S1(config-if)# switchport trunk allowed vlan 20,30,40,1000
Site1-S1# show interfaces trunk
```

```
Site1-S1(config)# interface range f0/3, f0/5
Site1-S1(config-if-range)# switchport mode trunk
Site1-S1(config-if-range)# switchport trunk native vlan 1000
Site1-S1(config-if-range)# switchport trunk allowed vlan 20,30,40,1000
```

```
Site1-S1(config)# spanning-tree portfast bpduguard default
Site1-S1(config)# interface F0/3
Site1-S1(config-if)# no spanning-tree portfast
Site1-S1(config-if)# no spanning-tree bpduguard
```

```
Site1-S1(config)# username admin secret cisco12345
Site1-S1(config)# ip domain name hsk-spare.com
Site1-S1(config)# crypto key generate rsa general-keys modulus 2048
Site1-S1(config)# line vty 0 4
Site1-S1(config-line)# transport input ssh
Site1-S1(config-line)# login local
Site1-S1(config-line)# exit
```

```
Site1-S1(config)# ntp server 192.168.5.253 prefer
Site1-S1(config)# exit
```

```
Site1-S1# copy running-config startup-config
```

### Site1-S2

```
switch(config)# hostname Site1-S2
Site1-S2(config)# no ip domain lookup
Site1-S2(config)# enable secret class
Site1-S2(config)# line console 0
Site1-S2(config-line)# password cisco
Site1-S2(config-line)# login
Site1-S2(config)# line vty 0 15
Site1-S2(config-line)# password cisco
Site1-S2(config-line)# login
Site1-S2(config)# service password-encryption
Site1-S2(config)# banner motd $ Unauthorized access is Prohibited! $
Site1-S2(config)# exit
```

```
Site1-S2(config)# vlan 20
Site1-S2(config-vlan)# name Management
Site1-S2(config-vlan)# vlan 30
Site1-S2(config-vlan)# name Operations
Site1-S2(config-vlan)# vlan 40
Site1-S2(config-vlan)# name Marketing
Site1-S2(config-vlan)# vlan 999
Site1-S2(config-vlan)# name ParkingLot
Site1-S2(config-vlan)# vlan 1000
Site1-S2(config-vlan)# name Native
Site1-S2(config-vlan)# exit
```

```
Site1-S2(config)# interface vlan 40
```

```
Site1-S2(config-if)# ip address dhcp
Site1-S2(config-if)# no shutdown
Site1-S2(config-if)# exit
Site1-S2(config)# ip default-gateway 192.168.1.1
Site1-S2(config)# interface range f0/5-24, g0/1-2
Site1-S2(config-if-range)# switchport mode access
Site1-S2(config-if-range)# switchport access vlan 999
Site1-S2(config-if-range)# shutdown
Site1-S2(config-if-range)# exit

Site1-S2(config)# interface range f0/1-2
Site1-S2(config-if-range)# channel-group 1 mode active
Site1-S2(config-if-range)# exit
Site1-S2(config)# interface port-channel 1
Site1-S2(config-if)# switchport mode trunk
Site1-S2(config-if)# switchport mode trunk native vlan 1000
Site1-S2(config-if)# switchport trunk allowed vlan 20,30,40,1000

Site1-S2(config)# interface range f0/3-4
Site1-S2(config-if-range)# channel-group 2 mode active
Site1-S2(config-if-range)# exit
Site1-S2(config)# interface port-channel 2
Site1-S2(config-if)# switchport mode trunk
Site1-S2(config-if)# switchport mode trunk native vlan 1000
Site1-S2(config-if)# switchport trunk allowed vlan 20,30,40,1000

Site1-S2(config)# spanning-tree portfast bpduguard default
Site1-S2(config)# username admin secret cisco12345
Site1-S2(config)# ip domain name hsk-spare.com
Site1-S2(config)# crypto key generate rsa general-keys modulus 2048
Site1-S2(config)# line vty 0 4
Site1-S2(config-line)# transport input ssh
Site1-S2(config-line)# login local
Site1-S2(config-line)# exit

Site1-S2(config)# ntp server 192.168.5.253 prefer
Site1-S2(config)# exit

Site1-S2# copy running-config startup-config
```

### Site1-S3

```
switch(config)# hostname Site1-S3
Site1-S3(config)# no ip domain lookup
Site1-S3(config)# enable secret class
Site1-S3(config)# line console 0
Site1-S3(config-line)# password cisco
Site1-S3(config-line)# login
Site1-S3(config)# line vty 0 15
Site1-S3(config-line)# password cisco
Site1-S3(config-line)# login
Site1-S3(config)# service password-encryption
Site1-S3(config)# banner motd $ Unauthorized access is Prohibited! $
```

```
Site1-S3(config)# vlan 50
Site1-S3(config-vlan)# name DMZ
Site1-S3(config-vlan)# vlan 999
Site1-S3(config-vlan)# name ParkingLot
```

```
Site1-S3(config)# interface vlan 50
Site1-S3(config-if)# ip address 192.168.4.2 255.255.255.192
Site1-S3(config-if)# no shutdown
Site1-S3(config-if)# exit
Site1-S3(config)# ip default-gateway 192.168.4.1
```

```
Site1-S3(config)# interface range f0/1, f0/6
Site1-S3(config-if-range)# switchport mode access
Site1-S3(config-if-range)# switchport access vlan 50
Site1-S3(config-if-range)# exit
Site1-S3(config)# interface range f0/2 - 5, f0/7 - 24, g0/1 - 2
Site1-S3(config-if-range)# switchport mode access
Site1-S3(config-if-range)# switchport access vlan 999
Site1-S3(config-if-range)# shutdown
Site1-S3(config-if-range)# exit
```

```
Site1-S3(config)# spanning-tree portfast bpduguard default
```

```
Site1-S3(config)# username SSHadmin secret cisco12345
Site1-S3(config)# ip domain name hsk-spare.com
Site1-S3(config)# crypto key generate rsa general-keys modulus 2048
Site1-S3(config)# line vty 0 4
Site1-S3(config-line)# transport input ssh
Site1-S3(config-line)# login local
Site1-S3(config-line)# exit
```

```
Site1-S3(config)# ntp server 192.168.5.253
Site1-S3(config)# exit
```

```
Site1-S3# copy running-config startup-config
```

### **Site1-S4**

```
switch(config)# hostname Site1-S4
Site1-S4(config)# no ip domain lookup
Site1-S4(config)# enable secret class
Site1-S4(config)# line console 0
Site1-S4(config-line)# password cisco
Site1-S4(config-line)# login
Site1-S4(config)# line vty 0 15
Site1-S4(config-line)# password cisco
Site1-S4(config-line)# login
Site1-S4(config)# service password-encryption
Site1-S4(config)# banner motd $ Unauthorized access is Prohibited! $
```

```
Site1-S4(config)# vlan 20
Site1-S4(config-vlan)# name Management
Site1-S4(config-vlan)# vlan 30
Site1-S4(config-vlan)# name Operations
Site1-S4(config-vlan)# vlan 40
Site1-S4(config-vlan)# name Marketing
Site1-S4(config-vlan)# vlan 999
Site1-S4(config-vlan)# name ParkingLot
Site1-S4(config-vlan)# vlan 1000
Site1-S4(config-vlan)# name Native
Site1-S4(config-vlan)# exit
```

```
Site1-S4(config)# interface vlan 20
Site1-S4(config-if)# ip address dhcp
Site1-S4(config-if)# no shutdown
Site1-S4(config-if)# exit
Site1-S4(config)# ip default-gateway 192.168.3.1
```

```
Site1-S4(config)# interface range f0/6 - 17
Site1-S4(config-if-range)# switchport mode access
Site1-S4(config-if-range)# switchport access vlan 20
```

```
Site1-S4(config-if-range)# exit
Site1-S4(config)# interface range f0/4 - 5, f0/18 - 24, g0/1 - 2
Site1-S4(config-if-range)# switchport mode access
Site1-S4(config-if-range)# switchport access vlan 999
Site1-S4(config-if-range)# shutdown
Site1-S4(config-if-range)# exit
```

```
Site1-S4(config)# interface range f0/1-2
Site1-S4(config-if-range)# channel-group 2 mode active
Site1-S4(config-if-range)# exit
Site1-S4(config)# interface port-channel 2
Site1-S4(config-if)# switchport mode trunk
Site1-S4(config-if)# switchport mode trunk native vlan 1000
Site1-S4(config-if)# switchport trunk allowed vlan 20,30,40,1000
Site1-S4# show interfaces trunk
```

```
Site1-S1(config)# interface f0/3
Site1-S1(config-if)# switchport mode trunk
Site1-S1(config-if)# switchport trunk native vlan 1000
Site1-S1(config-if)# switchport trunk allowed vlan 20,30,40,1000
```

```
Site1-S4(config)# spanning-tree portfast bpduguard default
Site1-S4(config)# interface F0/3
Site1-S4(config-if)# no spanning-tree portfast
Site1-S4(config-if)# no spanning-tree bpduguard
```

```
Site1-S4(config)# username admin secret cisco12345
Site1-S4(config)# ip domain name hsk-spare.com
Site1-S4(config)# crypto key generate rsa general-keys modulus 2048
Site1-S4(config)# line vty 0 4
Site1-S4(config-line)# transport input ssh
Site1-S4(config-line)# login local
Site1-S4(config-line)# exit
```

```
Site1-S4(config)# ntp server 192.168.5.253 prefer
Site1-S4(config)# exit
```

```
Site1-S4# copy running-config startup-config
```

### **S1Annex-S5**

```
switch(config)# hostname S1Annex-S5
S1Annex-S5(config)# no ip domain lookup
S1Annex-S5(config)# enable secret class
S1Annex-S5(config)# line console 0
S1Annex-S5(config-line)# password cisco
S1Annex-S5(config-line)# login
S1Annex-S5(config)# line vty 0 15
S1Annex-S5(config-line)# password cisco
S1Annex-S5(config-line)# login
S1Annex-S5(config)# service password-encryption
S1Annex-S5(config)# banner motd $ Unauthorized access is Prohibited! $
```

```
S1Annex-S5(config)# vlan 60
S1Annex-S5(config-vlan)# name R&D
S1Annex-S5(config-vlan)# vlan 999
S1Annex-S5(config-vlan)# name ParkingLot
```

```
S1Annex-S5(config)# interface vlan 60
S1Annex-S5(config-if)# ip address 192.168.6.2 255.255.255.192
S1Annex-S5(config-if)# no shutdown
S1Annex-S5(config-if)# exit
S1Annex-S5(config)# ip default-gateway 192.168.6.1
```

```
S1Annex-S5(config)# interface range f0/5, f0/6-24
S1Annex-S5 (config-if-range)# switchport mode access
S1Annex-S5 (config-if-range)# switchport access vlan 60
S1Annex-S5 (config-if-range)# exit
```

```
S1Annex-S5(config)# interface range f0/1 – 4, g0/1 - 2
S1Annex-S5(config-if-range)# switchport mode access
S1Annex-S5(config-if-range)# switchport access vlan 999
S1Annex-S5(config-if-range)# shutdown
S1Annex-S5(config-if-range)# exit
```

```
S1Annex-S5(config)# spanning-tree portfast bpduguard default
```

```
S1Annex-S5(config)# username admin secret cisco12345
S1Annex-S5(config)# ip domain name hsk-spare.com
S1Annex-S5(config)# crypto key generate rsa general-keys modulus 2048
S1Annex-S5(config)# line vty 0 4
S1Annex-S5(config-line)# transport input ssh
S1Annex-S5(config-line)# login local
```

```
S1Annex-S5(config-line)# exit
```

```
S1Annex-S5(config)# ntp server 192.168.5.253 prefer
S1Annex-S5(config)# exit
```

```
S1Annex-S5# copy running-config startup-config
```

### Site2-S1

```
switch(config)# hostname Site2-S1
Site2-S1(config)# no ip domain lookup
Site2-S1(config)# ip domain-name example.com
Site2-S1(config)# security passwords min-length 10
Site2-S1(config)# username admon01 algorithm-type scrypt secret cisco12345
Site2-S1(config)# line console 0
Site2-S1(config-line)# login local
Site2-S1(config-line)# exec-timeout 5 0
Site2-S1(config-line)# logging synchronous
Site2-S1(config-line)# exit
Site2-S1(config)# crypto key generate rsa general-keys modulus 2048
Site2-S1(config)# line vty 0 15
Site2-S1(config-line)# login local
Site2-S1(config-line)# transport input ssh
Site2-S1(config-line)# exec-timeout 5 0
Site2-S1(config)# Enable algorithm-type scrypt secret class12345
Site2-S1(config)# banner motd $ Unauthorized access is Prohibited! $
```

```
Site2-S1(config)# vlan 10
Site2-S1(config-vlan)# name Management
Site2-S1(config-vlan)# vlan 50
Site2-S1(config-vlan)# name Business
Site2-S1(config-vlan)# vlan 2
Site2-S1(config-vlan)# name Native
Site2-S1(config-vlan)# exit
```

```
Site2-S1(config)# interface vlan 10
Site2-S1(config-if)# ip address 192.168.10.2 255.255.255.0
Site2-S1(config-if)# no shutdown
Site2-S1(config-if)# exit
Site2-S1(config)# ip default-gateway 192.168.10.1
```

```
Site2-S1(config)# interface range G0/1-2
Site2-S1(config-if-range)# channel-group 1 mode active
Site2-S1(config-if-range)# exit
Site2-S1(config)# interface port-channel 1
Site2-S1(config-if)# switchport mode trunk
Site2-S1(config-if)# switchport mode trunk native vlan 2
Site2-S1(config-if)# switchport trunk allowed vlan 2,10,50
Site2-S1# show interfaces trunk
```

```
Site2-S1(config)# interface range f0/1 - 20
Site2-S1(config-if-range)# switchport mode access
Site2-S1(config-if-range)# switchport access vlan 50
Site2-S1(config-if-range)# exit
Site2-S1(config)# interface range f0/21, 23 - 24
Site2-S1(config-if-range)# switchport mode access
Site2-S1(config-if-range)# switchport access vlan 10
Site2-S1(config-if-range)# exit
```

```
Site2-S1(config)# interface range f0/22
Site2-S1(config-if-range)# switchport mode trunk
Site2-S1(config-if-range)# switchport trunk native vlan 2
Site2-S1(config-if-range)# switchport trunk allowed vlan 2,10,50
```

```
Site2-S1(config)# spanning-tree portfast bpduguard default
```

```
Site2-S1(config)# ntp server 192.168.5.253 prefer
Site2-S1(config)# exit
```

```
Site2-S1# copy running-config startup-config
```

### Site2-S2

```
switch(config)# hostname Site2-S2
Site2-S1(config)# no ip domain lookup
Site2-S1(config)# ip domain-name example.com
Site2-S1(config)# security passwords min-length 10
Site2-S1(config)# username admon01 algorithm-type scrypt secret cisco12345
Site2-S1(config)# line console 0
Site2-S1(config-line)# login local
Site2-S1(config-line)# exec-timeout 5 0
Site2-S1(config-line)# logging synchronous
```

```
Site2-S1(config-line)# exit
Site2-S1(config)# crypto key generate rsa general-keys modulus 2048
Site2-S1(config)# line vty 0 15
Site2-S1(config-line)# login local
Site2-S1(config-line)# transport input ssh
Site2-S1(config-line)# exec-timeout 5 0
Site2-S1(config)# Enable algorithm-type scrypt secret class12345
Site2-S1(config)# banner motd $ Unauthorized access is Prohibited! $
```

```
Site2-S1(config)# vlan 10
Site2-S1(config-vlan)# name Management
Site2-S1(config-vlan)# vlan 50
Site2-S1(config-vlan)# name Business
Site2-S1(config-vlan)# vlan 2
Site2-S1(config-vlan)# name Native
Site2-S1(config-vlan)# exit
```

```
Site2-S1(config)# interface vlan 10
Site2-S1(config-if)# ip address 192.168.10.3 255.255.255.0
Site2-S1(config-if)# no shutdown
Site2-S1(config-if)# exit
Site2-S1(config)# ip default-gateway 192.168.10.1
```

```
Site2-S2(config)# interface range G0/1-2
Site2-S2(config-if-range)# channel-group 1 mode active
Site2-S2(config-if-range)# exit
Site2-S2(config)# interface port-channel 1
Site2-S2(config-if)# switchport mode trunk
Site2-S2(config-if)# switchport mode trunk native vlan 2
Site2-S2(config-if)# switchport trunk allowed vlan 2,10,50
Site2-S2# show interfaces trunk
```

```
Site2-S2(config)# interface f0/1
Site2-S2(config-if)# switchport mode access
Site2-S2(config-if)# switchport access vlan 10
Site2-S2(config-if)# exit
Site2-S2(config)# interface range f0/2 - 24
Site2-S2(config-if-range)# switchport mode access
Site2-S2(config-if-range)# switchport access vlan 50
Site2-S2(config-if-range)# exit
```

```
Site2-S2(config)# spanning-tree portfast bpduguard default
```

```
Site2-S2(config)# ntp server 192.168.5.253 prefer
```

```
Site2-S2(config)# exit
```

```
Site2-S2# copy running-config startup-config
```